AskNearby: An LLM-Based Application for Neighborhood Information Retrieval and Personalized Cognitive-Map Recommendations

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Abstract

The "15-minute city" envisions neighborhoods where residents can meet daily needs via a short walk or bike ride. Realizing this vision requires not only physical proximity but also efficient and reliable access to information about nearby places, services, and events. Existing location-based systems, however, focus mainly on city-level tasks and neglect the spatial, temporal, and cognitive factors that shape localized decision-making. We conceptualize this gap as the Local Life Information Accessibility (LLIA) problem and introduce *AskNearby*, an AI-driven community application that unifies retrieval and recommendation within the 15-minute life circle. *AskNearby* integrates (i) a three-layer Retrieval-Augmented Generation (RAG) pipeline that synergizes graph-based, semantic-vector, and geographic retrieval with (ii) a cognitive-map model that encodes each user's neighborhood familiarity and preferences. Experiments on real-world community datasets demonstrate that *AskNearby* significantly outperforms LLM-based and map-based baselines in retrieval accuracy and recommendation quality, achieving robust performance in spatiotemporal grounding and cognitive-aware ranking. Real-world deployments further validate its effectiveness. By addressing the LLIA challenge, *AskNearby* empowers residents to more effectively discover local resources, plan daily activities, and engage in community life.

CCS Concepts

• **Information systems** → **Location based services**; **Information retrieval**; • **Computing methodologies** → *Natural language processing*; *Planning and scheduling*.

Keywords

Large Language Models, Retrieval-Augmented Generation, Spatiotemporal Knowledge Graph, Local Information, 15-Minute City

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1 Introduction

The 15-minute city concept envisions that people can fulfill basic daily needs—such as grocery shopping, healthcare, education, and leisure—within a short walking or cycling distance from their homes, typically within 15 minutes [17]. This idea has gained widespread attention among urban planners and policymakers, who increasingly emphasize proximity-based planning and active transportation as essential principles for promoting sustainable urban development. In parallel with this, researchers have conducted empirical studies on whether residents can reach nearby destinations and access urban functions [1, 4, 12, 24].

Beyond physical accessibility, digitalization represents another core principle of the 15-minute city. It entails leveraging smart city technologies—such as big data and the Internet of Things—to support real-time information delivery, citizen engagement, and efficient urban resource management [11]. However, digital infrastructure alone does not guarantee that individuals can effectively access or utilize localized information [5]. This raises a critical but largely unaddressed question: Can people access relevant information about nearby places and services to make informed decisions in

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AskNearby：一个基于大型语言模型的邻里信息检索和个性化认知地图推荐应用

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摘要

“15分钟城市”设想的是居民可以通过短途步行或骑行满足日常需求的社区。实现这一愿景不仅需要物理上的接近，还需要高效可靠地获取有关附近地点、服务和活动的信息。然而，现有的基于位置的系统主要关注城市级别的任务，忽略了影响本地化决策的空间、时间和认知因素。我们将这一差距概念化为本地生活信息可达性（LLIA）问题，并推出了 AskNearby，这是一款 AI 驱动的社区应用，在 15 分钟生活圈内整合了检索与推荐。AskNearby 集成了 (i) 三层检索增强生成（RAG）流程，将基于图的、语义向量的和地理检索协同起来，以及 (ii) 编码每个用户对社区熟悉度和偏好的认知地图模型。在现实社区数据集上的实验表明，AskNearby 显著

CCS 概念

• 信息系统 → 基于位置的服务；信息检索；
• 计算方法 → 自然语言处理；计划与调度。

关键词

大语言模型、检索增强生成、时空知识图谱、本地信息、15分钟城市

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1 引言

15分钟城市的概念设想，人们可以在离家很近的步行或骑行距离内满足基本的日常需求——如购物、医疗、教育和休闲——通常在15分钟内[17]。这一理念在城市规划者和政策制定者中引起了广泛关注，他们越来越强调基于邻近性的规划和积极交通作为促进可持续城市发展的基本原则。与此相应，研究人员也进行了实证研究，以探讨居民是否能够到达附近的目的地并获取城市功能[1, 4, 12, 24]。

除了物理可达性之外，数字化也是“15分钟城市”的另一个核心原则。它涉及利用智慧城市技术——如大数据和物联网——来支持实时信息传递、市民参与以及高效的城市资源管理[11]。然而，仅有数字基础设施并不能保证个人能够有效访问或利用本地化信息[5]。这提出了一个关键但在很大程度上未被解决的问题：人们是否能够获取关于附近地点和服务的相关信息，以便做出明智的决策

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their daily lives, especially in dynamic environments where places, facilities, and human activities are constantly changing?

Existing applications are ill-equipped to address this need of accessing local information at a granular, neighborhood scale [3, 16]. Their failure stems from an inability to model the nuanced spatiotemporal context, ensuring that the offered information is both timely and locally relevant. To overcome this limitation, it is necessary to support two complementary processes through which users interact with local information: *active retrieval* and *passive recommendation*.

In active retrieval, users explicitly seek information regarding nearby services or events, often formulating queries inherently containing spatiotemporal constraints (e.g., 'Which supermarket near me is open late?'). Traditional search engines rely heavily on keyword-based matching and fixed boundaries, such as a predefined search radius or time windows, failing to grasp the complex spatial and semantic intent in human queries [34]. Recent studies have explored the potential of Large Language Models (LLMs) to interpret semantic and spatiotemporal constraints on local information and activities, but most approaches still treat space and time as isolated filters rather than as interdependent factors shaping local knowledge [14, 18, 27].

Passive recommendation proactively surfaces relevant information without explicit user queries. Achieving effective passive recommendation in a local context necessitates understanding not only individuals' preferences but also the broader spatial cognition that shapes their activity choices within a community. Although recommender systems have been extensively studied in urban computing [33], most existing approaches operate at a city-wide scale, relying on conventional collaborative filtering or content-based methods [3, 19]. These methods, however, often fail to accurately capture community-scale behavioral patterns or leverage residents' unique cognitive understanding of places. As a result, they struggle to provide localized recommendations that align with how people perceive and navigate their immediate environments.

To address these challenges, we define the problem of **Local Life Information Accessibility** (LLIA) as enabling residents to efficiently obtain relevant, timely, and context-aware local information through both active search and passive recommendation. Distinct from traditional search or recommendation tasks, LLIA requires jointly modeling spatial proximity, temporal relevance, and cognitive familiarity at the neighborhood scale, which poses unique technical and representational challenges. To support LLIA, we propose *AskNearby*, an AI-driven community system built on a unified spatiotemporal retrieval and recommendation framework powered by LLMs. *AskNearby* consists of two main components: First, a three-layer Retrieval-Augmented-Generation (RAG) architecture that facilitates active information seeking by synergizing **GraphRAG** (graph-based retrieval), **VectorRAG** (semantic similarity retrieval), and **GeoRAG** (geographic-based retrieval); second, a cognitive map model that captures users' spatial preferences to support passive, location-aware personalized recommendations.

Our main contributions are threefold:

- We define and formalize the LLIA problem, which encompasses both active search and passive recommendation to support spatiotemporal, context-aware access to neighborhood-scale information.
- We design and develop *AskNearby*, a novel community platform that integrates a multi-layered LLM-based RAG architecture to enhance local information retrieval and a cognitive map-based recommendation system.
- We validate *AskNearby* through extensive evaluations of *AskNearby* on real-world local community datasets, demonstrating its superior performance in providing more accurate and context-aware local information compared to baseline methods.

2 Related Works

This section reviews prior research on spatiotemporal information retrieval and recommendation, with a particular focus on methods that leverage textual data, natural language queries, and user-generated content. We begin by reviewing traditional spatial and temporal approaches, and subsequently, we discuss recent advances driven by LLMs.

2.1 Local Information Retrieval and Recommendation

Recent work on local information retrieval and recommendation primarily addresses the spatial properties of Points of Interest (POIs) and the temporal signals captured in user-generated or activity-related content, such as check-ins, textual reviews, and event listings. A range of techniques have been developed to improve retrieval accuracy and recommendation efficiency. Representative approaches include context-aware modeling of user behavior [2, 31], spatial indexing and query optimization [15, 16], time-sensitive recommendation strategies [3, 20], and hybrid systems that combine content-based and collaborative filtering [26].

However, existing approaches tend to focus on retrieving structured, global-scale content rather than localized, contextualized knowledge. This leads to two key limitations. First, users often need to interpret search results to extract meaningful local insights manually. Second, most systems are not designed to handle open-domain, natural-language queries about local life—queries that are inherently ambiguous, context-rich, and highly personalized.

2.2 LLMs for Spatiotemporal Information Retrieval and Recommendation

To address these challenges, recent work has turned to LLMs, which have shown strong capabilities in understanding natural language queries and reasoning over complex content [14]. Empirical studies demonstrate that LLMs can effectively process time-series data, geospatial inputs, and crowd-sourced textual content [7, 10]. They are also capable of inferring temporal relations from unstructured text [28], improving geolocation representations through language understanding [9], and identifying implicit spatial references embedded within user expressions [25]. These emergent capabilities position LLMs as highly promising tools for spatiotemporal information access, particularly in local contexts where spatial, temporal, and cognitive factors are deeply intertwined [14].

Capitalizing on these strengths, recent efforts have applied LLMs to spatiotemporal information retrieval and recommendation tasks.

他们的日常生活，特别是在那些地点、设施和人类活动不断变化的动态环境中?

现有的应用程序无法有效满足以细粒度、社区级别访问本地信息的需求 [3, 16]。其失败源于无法建模细微的时空背景，确保所提供的信息既及时又具有本地相关性。为克服这一限制，有必要通过两种互补的过程支持用户与本地信息的交互：主动检索和被动推荐。

在主动检索中，用户会明确地寻找有关附近服务或活动的信息，通常会形成天生包含时空限制的查询（例如，“我附近哪个超市营业到很晚？”）。传统搜索引擎在很大程度上依赖基于关键词的匹配和固定边界，例如预定义的搜索半径或时间窗口，无法理解人类查询中的复杂空间和语义意图[34]。最近的研究探讨了大型语言模型（LLM）在解读本地信息和活动的语义及时空限制方面的潜力，但大多数方法仍然将空间和时间视为孤立的过滤条件，而不是作为塑造本地知识的相互依赖因素[14, 18, 27]。

被动推荐能够在没有明确用户查询的情况下主动呈现相关信息。在本地环境中实现有效的被动推荐不仅需要理解个体的偏好，还需要理解塑造他们在社区内活动选择的更广泛空间认知。尽管推荐系统在城市计算中已经被广泛研究 [33]，但大多数现有方法是在城市范围内操作，依赖于传统的协同过滤或基于内容的方法 [3, 19]。然而，这些方法经常无法准确捕捉社区规模的行为模式，也无法利用居民对地点的独特认知。因此，它们难以提供与人们感知和导航其周围环境方式相一致的本地化推荐。

为应对这些挑战，我们将本地生活信息可获取性（Local Life Information Accessibility, LLIA）的问题定义为使居民能够通过主动搜索和被动推荐，高效获取相关、及时且具有情境感知的本地信息。不同于传统的搜索或推荐任务，LLIA 需要在社区尺度上联合建模空间接近性、时间相关性和认知熟悉度，这带来了独特的技术和表示挑战。为支持 LLIA，我们提出了 AskNearby，这是一个基于统一时空检索与推荐框架、由大型语言模型（LLMs）驱动的社区人工智能系统。AskNearby 由两个主要组成部分构成：首先，一个三层检索增强生成（Retrieval-Augmented-Generation, RAG）架构，通过整合 GraphRAG（基于图的检索）、VectorRAG（语义相似性检索）和 GeoRAG（基于地理的检索），促进主动信息获取；其次，一个认知地图模型，用于捕捉用户的

我们的主要贡献有三方面：

- 我们定义并形式化了LLIA问题，该问题涵盖主动搜索和被动推荐
支持时空、上下文感知的邻域访问 -
比例信息。
- 我们设计并开发了 AskNearby，这是一个新型社区平台，集成了多层次的基于大语言模型的 RAG 架构，以增强本地信息检索，并结合基于认知地图的推荐系统。
- 我们通过在真实世界的本地社区数据集上对 AskNearby 进行广泛评估来验证 AskNearby，证明其在提供比基线方法更准确和具有上下文感知的本地信息方面表现出更优的性能。

2 相关工作

本节回顾了关于时空信息检索和推荐的先前研究，特别关注利用文本数据、自然语言查询以及用户生成内容的方法。我们首先回顾传统的空间和时间方法，随后讨论由大语言模型推动的最新进展。

2.1 本地信息检索与推荐

近期关于本地信息检索和推荐的研究主要涉及兴趣点（POIs）的空间特性以及用户生成或与活动相关的内容中捕捉到的时间信号，例如签到、文本评论和活动列表。为提高检索准确性和推荐效率，已经开发了多种技术。具有代表性的方法包括用户行为的上下文感知建模 [2, 31]、空间索引和查询优化 [15, 16]、时间敏感的推荐策略 [3, 20]，以及结合基于内容和协同过滤的混合系统 [26]。

然而，现有的方法往往侧重于检索结构化的、全球范围的内容，而不是局部的、情境化的知识。这导致了两个关键的限制。首先，用户通常需要手动解释搜索结果以提取有意义的局部见解。其次，大多数系统并未设计用于处理关于本地生活的开放领域自然语言查询——这些查询本质上具有歧义、情境丰富且高度个性化。

2.2 用于时空信息检索与推荐的大型语言模型

为了解决这些挑战，最近的研究转向了大型语言模型（LLM），它们在理解自然语言查询和推理复杂内容方面显示出了强大的能力 [14]。实证研究表明，LLM 可以有效处理时间序列数据、地理空间输入和众包文本内容 [7, 10]。它们还能够从非结构化文本中推断时间关系 [28]，通过语言理解提升地理位置信息表示 [9]，并识别嵌入在用户表达中的隐性空间引用 [25]。这些新兴能力使 LLM 成为获取时空信息的极具前景的工具，特别是在空间、时间和认知因素密切交织的本地环境中 [14]。

利用这些优势，最近的努力已经将大型语言模型应用于时空信息检索和推荐任务。

For example, EvoRAG integrates retrieved user trajectories with LLM reasoning to support personalized travel planning [**?**]. Tian et al. [21] present a retrieval and re-ranking framework that uses LLMs to identify similar environmental events from web and news sources. Li et al. [13] propose an LLM-based framework for next POI recommendation that preserves rich contextual signals from location-based social network data, effectively addressing challenges such as cold-start and short user trajectories. LLMs have also exhibited potential in interpreting nuanced spatial descriptions (e.g., "nearby", "within walking distance") [8] and temporal phrases (e.g., "open tonight", "happening this weekend") [29], which traditional systems often struggle to handle.

Despite these notable advances, current LLM-based applications typically lack a holistic spatiotemporal perspective [22, 32]. Such systems often prioritize textual semantics while overlooking the complex interplay of time, space, user context, and cognitive spatial familiarity. Consequently, significant challenges remain in delivering user-centric, context-aware, and personalized access to local life information—especially within the dynamic, fine-grained environments envisioned by the 15-minute city paradigm.

3 Methodology

3.1 Problem Definition: LLIA

LLIA refers to the ability of residents to efficiently obtain timely and context-aware local information. It encompasses two complementary processes: active retrieval, which is based on explicit user queries, and passive recommendation, which is driven by users' spatial familiarity and local preference patterns.

Formally, a user u is characterized with spatial position (s_u), temporal context (t_u), and frequently visited places (p_u), which approximate cognitive preferences. Given a spatiotemporal knowledge base K (containing POIs, events, and user-generated content with geographic, temporal, and semantic metadata), LLIA aims to return a set of relevant items $I = \{i_1, \ldots, i_n\}$ that are both recent and proximate. The LLIA function (A) can be defined as:

$$A(u) = \underbrace{\text{Retrieve}(Q, s_u, K)}_{\text{Active retrieval}} \cup \underbrace{\text{Recommend}(s_u, t_u, p_u, K)}_{\text{Passive recommendation}}. \quad (1)$$

The $\text{Retrieve}(Q, s_u, K)$ module retrieves candidate items relevant to the user's query Q and current location s_u. The $\text{Recommend}(s_u, t_u, p_u, K)$ module further ranks or selects results based on cognitive relevance, incorporating semantic similarity, spatial proximity, and public familiarity derived from s_u, t_u, and p_u.

3.2 System Overview

To enhance LLIA and provide local knowledge in a user-centric manner, we propose *AskNearby*, an AI-driven community system. An overview of this system is illustrated in Figure 1. *AskNearby* combines a multi-layered RAG architecture with a cognitive map model that captures users' spatial preferences. This unified framework enables both active information retrieval and passive, personalized recommendations based on spatial familiarity. Lastly, it provides an interactive user interface designed to facilitate real-time question-answering, personalized recommendation posts, and map-based visualization of nearby local information, as demonstrated in Figure 5 of Appendix A.

3.3 RAGs for Local Information Access

To support open-domain, spatiotemporal local queries, we employ a multi-layer RAG framework that decomposes and enriches the user query via a structured processing pipeline. Upon receiving a user-issued query Q, an LLM is initially utilized to extract spatial entities (e.g., "Futian Exhibition Center") and semantic intents (e.g., "restaurants", "entertainment") from the input.

Subsequently, a geospatial agent is invoked to query external APIs (e.g., Gaode Maps) and construct geometric representations of the identified locations in the form of points, polylines, or polygons. These structured spatial objects, together with the user's current location, are passed to GeoRAG to retrieve geographically proximate candidates.

In parallel, the semantic intent extracted from the query is used to direct GraphRAG, which retrieves conceptually or relationally relevant entities. Finally, the original query Q is concatenated with the graph-expanded semantic results and sent to VectorRAG to retrieve semantically similar posts or documents in the vector space.

The overall retrieval pipeline is expressed as:

$$\begin{aligned}\text{Retrieve}(Q, s_u) = {} & \text{GeoRAG}(\text{Loc}_Q, s_u) \\ & + \text{GraphRAG}(\text{Sem}_Q) \\ & + \text{VectorRAG}(Q \oplus \text{Sem}_G),\end{aligned} \quad (2)$$

where Loc_Q and Sem_Q denote location and semantic intent extracted from Q, and Sem_G is the graph-augmented semantic context.

GeoRAG focuses on accurate spatial computation and location-based filtering. Backed by a spatial database (e.g., *PostGIS*[1]), it supports geo-indexing, proximity querying, and spatial joins. GeoRAG ensures that search results are geographically constrained and contextually relevant, such as identifying facilities within a specific walking distance or filtering nearby services based on real-time user location. Given a query location Loc_Q (if available) or the user's current location s_u, and a candidate set of spatial entities S that filtered from the spatiotemporal knowledge base K, GeoRAG retrieves:

$$\text{GeoRAG}(\text{Loc}_Q, s_u, S) = \{s \in S \mid \text{dist}(\text{Loc}, s) < \theta\}, \quad (3)$$

where $\text{Loc} = \text{Loc}_Q$ if provided, otherwise $\text{Loc} = s_u$. Here, $\text{dist}(\cdot)$ denotes a spatial distance function (e.g., haversine), and θ is a predefined threshold (e.g., 1km).

GraphRAG leverages semantic graphs to retrieve conceptually and relationally relevant information. Built on a graph database (e.g., *NebulaGraph*[2]), it supports efficient traversal, neighborhood expansion, and relation-based filtering. The semantic graph encodes relationships between entities such as POI categories, tags, events, or user interests. Given a query intent Sem_Q extracted from the user input, GraphRAG identifies semantically related nodes and paths based on graph connectivity and relation types. For a graph G—constructed based on the information from K—and query intent Sem_Q, the semantic expansion can be defined as:

$$\text{GraphRAG}(\text{Sem}_Q, G) = \{e \in G \mid \text{rel}(\text{Sem}_Q, e) \in R\}, \quad (4)$$

[1] https://postgis.net/
[2] https://www.nebula-graph.com.cn/

例如，EvoRAG 将检索到的用户轨迹与大语言模型（LLM）的推理结合起来，以支持个性化的旅行规划 [?]。Tian 等人 [21] 提出了一个检索与重排序框架，使用 LLM 从网络和新闻源中识别相似的环境事件。Li 等人 [13] 提出了一种基于 LLM 的下一兴趣点推荐框架，该框架保留了基于位置的社交网络数据中的丰富上下文信号，有效地解决了冷启动和用户轨迹短等挑战。LLM 在解释细微的空间描述（例如，“附近”，“步行可达”）[8] 和时间短语（例如，“今晚开放”，“本周末发生”）[29] 方面也展示了潜力，而传统系统往往难以处理这些情况。

尽管有这些显著的进展，目前基于大语言模型（LLM）的应用通常缺乏整体的时空视角[22, 32]。此类系统往往优先考虑文本语义，而忽视时间、空间、用户情境和认知空间熟悉度的复杂相互作用。因此，在提供以用户为中心、情境感知和个性化的本地生活信息方面仍存在重大挑战——尤其是在“15分钟城市”范式所设想的动态、精细化环境中。

3 方法论

3.1 问题定义：LLIA

LLIA 指居民高效获取及时且具情境感知的本地信息的能力。它包括两个互补的过程：主动检索，即基于明确的用户查询，以及被动推荐，即由用户的空间熟悉度和本地偏好模式驱动。

形式上，用户 u 的特征包括空间位置 (s_u)、时间上下文 (t_u) 和经常访问的地点 (p_u)，这些特征近似反映了认知偏好。给定一个包含具有地理、时间和语义元数据的兴趣点、事件和用户生成内容的时空知识库 K ()，LLIA 旨在返回一组既近期又邻近的相关项目 $I = \{i_1, \ldots, i_n\}$。LLIA 函数 (A) 可以定义为：

$$A(u) = \underbrace{\text{Retrieve}(Q, s_u, K)}_{\text{Active retrieval}} \cup \underbrace{\text{Recommend}(s_u, t_u, p_u, K)}_{\text{Passive recommendation}}. \quad (1)$$

Retrieve(Q, s_u, K) 模块检索与用户查询 Q 和当前位置信息 s_u 相关的候选项目。Recommend(s_u, t_u, p_u, K) 模块则基于认知相关性进一步对结果进行排序或选择，结合从 s_u、t_u 和 p_u 得到的语义相似性、空间接近性和公众熟悉度。

3.2 系统概述

为了增强 LLIA 并以以用户为中心的方式提供本地知识，我们提出了 AskNearby，一种由 AI 驱动的社区系统。该系统的概览如图 1 所示。AskNearby 结合了多层 RAG 架构与认知地图模型，该模型捕捉用户的空间偏好。这个统一框架既支持主动的信息检索，也支持基于空间熟悉度的被动个性化推荐。最后，它提供了一个交互式用户界面，旨在促进实时问答、个性化推荐帖子以及基于地图的附近本地信息可视化，如附录 A 的图 5 所示。

3.3 用于本地信息访问的RAG

为了支持开放域的时空本地查询，我们采用了一个多层RAG框架，通过结构化处理流程分解并丰富用户查询。在接收到用户发出的查询 Q 后，首先使用大型语言模型（LLM）从输入中提取空间实体（例如，“福田会展中心”）和语义意图（例如，“餐厅”、“娱乐”）。

随后，调用地理空间代理查询外部 API（例如，高德地图），并将识别出的地点构建为点、折线或多边形形式的几何表示。这些结构化的空间对象连同用户的当前位置一起传递给 GeoRAG，以检索地理上接近的候选项。

同时，从查询中提取的语义意图用于指导 GraphRAG，该工具检索概念上或关系上相关的实体。最后，将原始查询 Q 与图扩展的语义结果连接起来，并发送到 VectorRAG，以在向量空间中检索语义上相似的帖子或文档。

整体检索流程表示为：

$$\begin{aligned}\text{Retrieve}(Q, s_u) = {} & \text{GeoRAG}(\text{Loc}_Q, s_u) \\ & + \text{GraphRAG}(\text{Sem}_Q) \\ & + \text{VectorRAG}(Q \oplus \text{Sem}_G),\end{aligned} \quad (2)$$

其中 Loc_Q 和 Sem_Q 分别表示从 Q 提取的位置和语义意图，Sem_G 是图增强语义上下文。

GeoRAG 专注于精准的空间计算和基于位置的过滤。依托空间数据库（例如 PostGIS[1]），它支持地理索引、邻近查询和空间连接。GeoRAG 确保搜索结果在地理上受到约束且具有上下文相关性，例如识别特定步行距离内的设施或根据实时用户位置过滤附近服务。给定查询位置 Loc_Q （（如果可用））或用户当前所在位置 s_u，以及从时空知识库 K 中筛选出的候选空间实体集合 S，GeoRAG 会检索：

$$\text{GeoRAG}(\text{Loc}_Q, s_u, S) = \{s \in S \mid \text{dist}(\text{Loc}, s) < \theta\}, \quad (3)$$

其中 $\text{Loc} = \text{Loc}_Q$ 如果提供，否则为 $\text{Loc} = s_u$。这里，dist$(\cdot)$ 表示空间距离函数（例如，haversine），而 θ 是预定义的阈值（例如，1公里）。

GraphRAG 利用语义图来检索概念上和关系上相关的信息。基于图数据库（例如 NebulaGraph[2]）构建，它支持高效的遍历、邻域扩展和基于关系的过滤。语义图编码了实体之间的关系，例如 POI 类别、标签、事件或用户兴趣。给定从用户输入中提取的查询意图 Sem_Q，GraphRAG 基于图的连通性和关系类型识别语义相关的节点和路径。对于基于 K 的信息构建的图 G 和查询意图 Sem_Q，语义扩展可以定义为：

$$\text{GraphRAG}(\text{Sem}_Q, G) = \{e \in G \mid \text{rel}(\text{Sem}_Q, e) \in R\}, \quad (4)$$

[1] https://postgis.net/
[2] https://www.nebula-graph.com.cn/

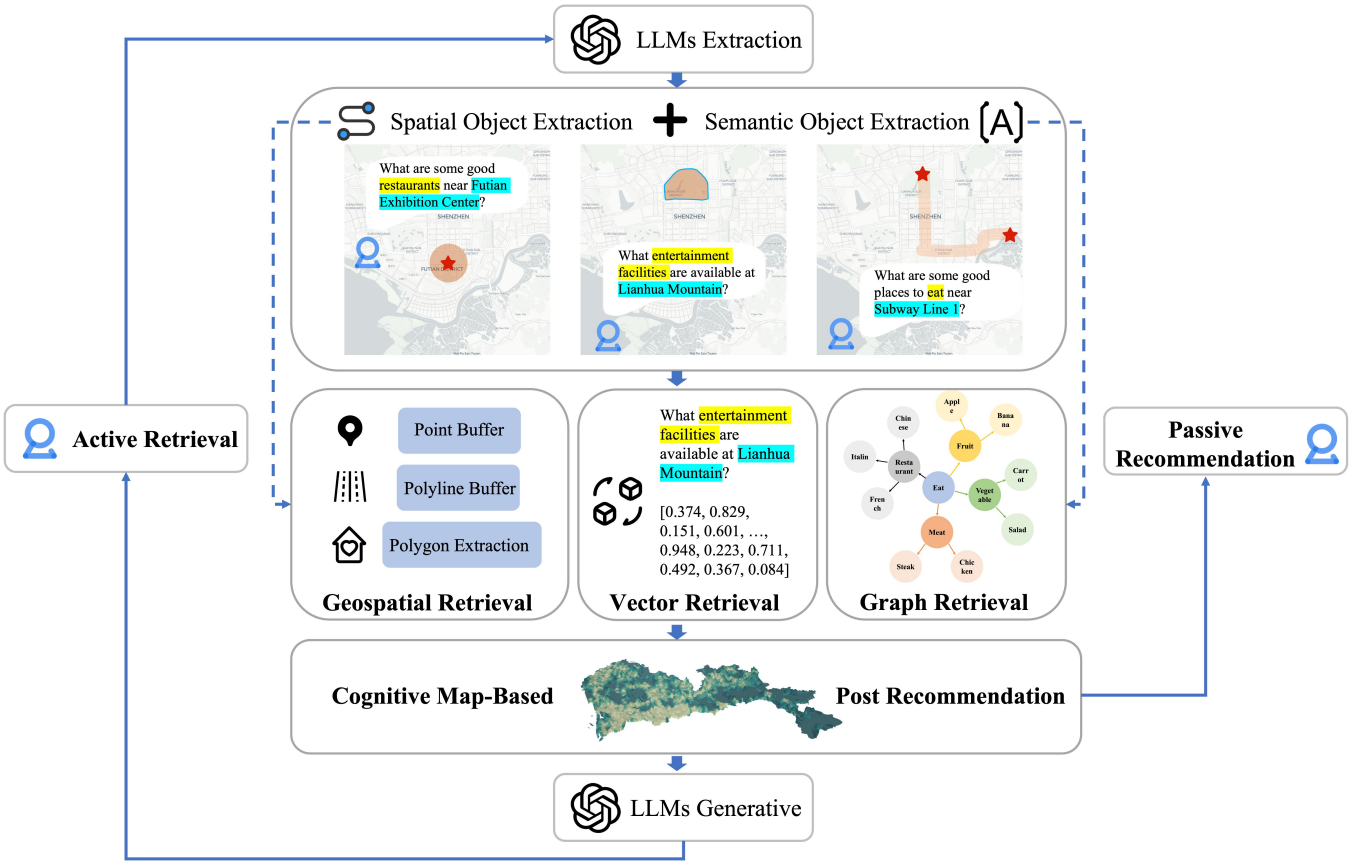

Figure 1: System overview of *AskNearby*, combining LLM-based object extraction, multi-source retrieval (geospatial, vector, graph), and a cognitive map model to support active information retrieval and passive local recommendations.

where R denotes a set of predefined semantic or relational edges in the graph, and $\text{rel}(\cdot)$ indicates the presence of a meaningful semantic connection.

VectorRAG enables semantic-level retrieval by supporting approximate nearest neighbor search over candidate contents. VectorRAG refines results from the outputs of GeoRAG and GraphRAG by ranking them based on semantic similarity to the enhanced query. Specifically, we concatenate the original query Q with graph-augmented semantic context Sem_G to form an enriched input $Q' = Q \oplus \text{Sem}_G$.

We utilize *pgvector*[3] as the vector similarity search engine, allowing efficient storage and querying of dense embeddings within a PostgreSQL-compatible environment. Both user queries and local content are embedded into high-dimensional vector representations through LLM-based encoders. Specifically, we adopt the *bge-large-zh-v1.5* model released by the Beijing Academy of Artificial Intelligence [23], which has demonstrated strong performance in semantic understanding tasks for Chinese text. We use Euclidean distance as the similarity metric to retrieve the most semantically relevant results from the filtered candidate set V' composed of these outputs from GeoRAG and GraphRAG. Formally, with encoder $\phi(\cdot)$ and candidate set V' derived from previous modules:

$$\text{VectorRAG}(Q, V') = \{v \in V' \mid \text{sim}\left(\phi(Q \oplus \text{Sem}_G), \phi(v)\right) > \delta\}, \tag{5}$$

where the function $\text{sim}(\cdot)$ denotes a similarity measure derived as the Euclidean distance between embeddings, and the threshold δ controls the minimum similarity required for a candidate v to be selected.

Beyond active retrieval, personalized and context-aware recommendations are also essential for enhancing accessibility to local life information. To this end, we introduce a cognitive map model that captures individual user preferences across space and time, as described in the following subsection.

3.4 Cognitive Map Model for Personalized Recommendation

To support contextualized and personalized recommendations in local life scenarios, we propose a cognitive map model that captures how users internally perceive and associate meaning with urban spaces over time. Users' preferences vary not only by spatial proximity but also by how they functionally and socially interpret different places. To translate these human-centered cognitive preferences into actionable recommendations, we define the recommendation

[3]https://github.com/pgvector/pgvector

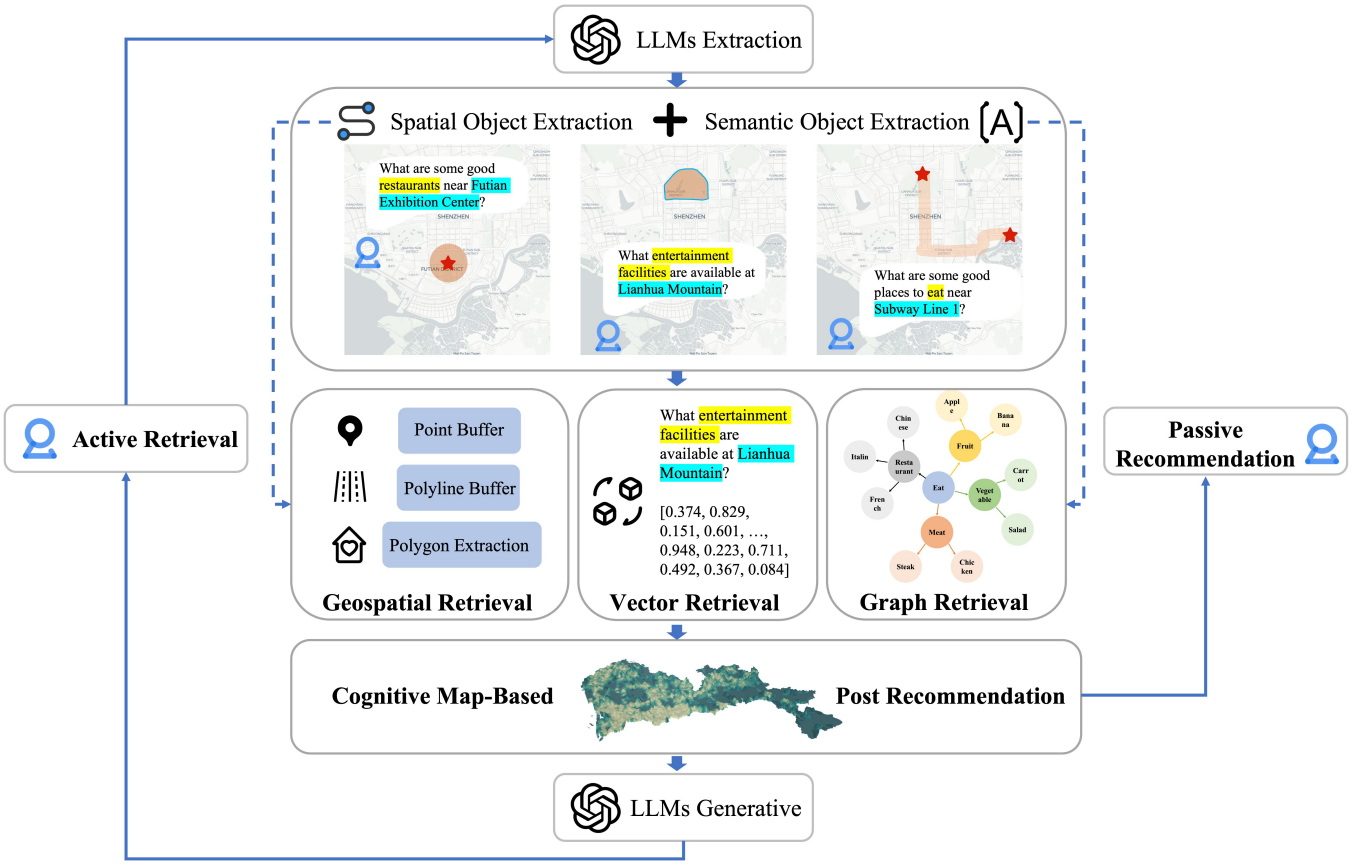

图 1：AskNearby 系统概览，结合基于大型语言模型的对象提取、多源检索（地理空间、向量或者，图），以及一个认知地图模型以支持主动信息检索和被动本地推荐。

其中 R 表示图中预定义的语义或关系边的集合，而 rel(·) 表示存在有意义的语义连接。

VectorRAG 通过支持对候选内容进行近似最近邻搜索，实现了语义级别的检索。VectorRAG 通过根据与增强查询的语义相似性对 GeoRAG 和 GraphRAG 的输出结果进行排序，从而优化结果。具体而言，我们将原始查询 Q 与图增强的语义上下文 Sem_G 连接起来，形成一个丰富的输入 $Q' = Q \oplus \text{Sem}_G$。

我们使用 pgvector[3] 作为向量相似度搜索引擎，从而在兼容 PostgreSQL 的环境中实现高效的密集向量存储和查询。用户查询和本地内容都通过基于大语言模型（LLM）的编码器嵌入为高维向量表示。具体来说，我们采用北京智源人工智能研究院发布的 bge-large-zh-v1.5 模型[23]，该模型在中文文本语义理解任务中表现出强大的性能。我们使用欧几里得距离作为相似度度量，从由 GeoRAG 和 GraphRAG 输出组成的过滤候选集合 V' 中检索最具语义相关性的结果。形式上，使用编码器 $\phi(\cdot)$ 以及由先前模块得出的候选集 V'：

$$\text{VectorRAG}(Q, V') = \{v \in V' \mid \text{sim}\left(\phi(Q \oplus \text{Sem}_G), \phi(v)\right) > \delta\}, \tag{5}$$

其中函数 sim(·) 表示由嵌入之间的欧几里得距离得到的相似度度量，阈值 δ 控制被选中候选 v 所需的最小相似度。

除了主动检索外，个性化和情境感知的推荐对于增强本地生活信息的可达性也至关重要。为此，我们引入了一种认知地图模型，该模型能够捕捉用户在空间和时间上的个体偏好，如以下小节所述。

3.4 个性化推荐的认知地图模型

为了支持本地生活场景中的情境化和个性化推荐，我们提出了一种认知地图模型，该模型捕捉了用户随时间对城市空间的内部感知和意义联想。用户的偏好不仅随空间接近性而变化，还取决于他们如何从功能和社会角度解读不同的地点。为了将这些以人为中心的认知偏好转化为可操作的推荐，我们定义了推荐

[3]https://github.com/pgvector/pgvector

score for an information item i as:

$$\Psi(s_u, t_u, p_u, i) = f_{\text{sem}}(s_u, t_u, p_u, i)^{\alpha} \cdot f_{\text{dist}}(d(s_u, i))^{\beta} \cdot f_{\text{pop}}(i)^{\gamma}, \quad (6)$$

where α, β, γ control the relative contributions of semantic relevance, spatial proximity, and public familiarity.

In this formulation, the overall recommendation score is computed as the product of three components: the **semantic relevance** term $f_{\text{sem}}(s_u, t_u, p_u, i)$, which measures the similarity between the user's cognitive profile and the temporal functional semantics of the item's location; the **spatial proximity** term $f_{\text{dist}}(d(s_u, i))$, which models geographic relevance via a distance-decay function based on the spatial distance between the user and the item; and the **public familiarity** term $f_{\text{pop}}(i)$, which reflects the aggregated popularity or visitation frequency of the location, capturing its collective cognitive salience.

To reflect the relative importance of different components in the recommendation model, we assign each factor an exponent α, β, and γ, which serve as soft weights in the multiplicative scoring function. We initialize these three weights equally at 1.0 to avoid imposing any prior preference, allowing semantic relevance, spatial proximity, and public familiarity to contribute equally to the final ranking. We acknowledge that a more optimal weight combination could potentially be found through grid search on the validation set, and we leave this for future work.

Figure 2 illustrates the conceptual model behind this scoring framework, showing how the semantic, spatial, and popularity signals interact to determine which content to recommend to the user in different contexts.

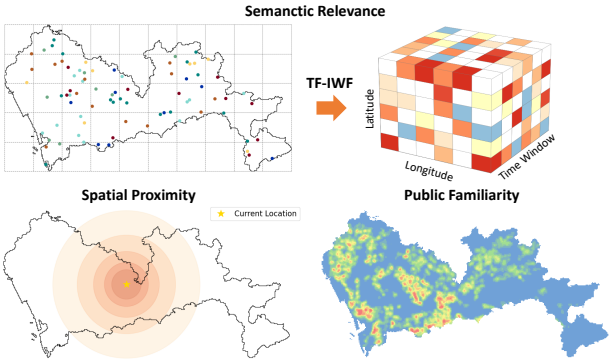

Figure 2: Illustration of the recommendation framework in *AskNearby*, combining semantic relevance, spatial proximity, and public familiarity to compute personalized scores.

Semantic relevance. To model how users cognitively perceive and associate meaning with places, we implement the semantic component using a TF-IWF (Term Frequency-Inverse Word Frequency) based representation [6], which captures the time-varying functional semantics of each location. This scheme identifies place attributes that are locally prominent yet globally distinctive across different time periods. Formally, the TF-IWF score is computed as:

$$\text{TF-IWF}_{k,g} = \frac{A_{k,g}}{\sum_i A_{i,g}} \cdot \log\left(\frac{\sum_h A_{k,h}}{\sum_i \sum_h A_{i,h}}\right), \quad (7)$$

where $A_{k,g}$ denotes the count of functional attribute k (e.g., restaurant, office, park) at location g during a specific time window. For example, a location may exhibit office-related semantics during the day and shift toward dining in the evening.

The semantic similarity $f_{\text{sem}}(s_u, t_u, p_u, i)$ is computed as the inner product between the TF-IWF vector of the item's location at time t_u and the user's cognitive profile. This profile is constructed by aggregating the TF-IWF vectors of the locations in p_u, the set of frequently visited places by user u. Each location's contribution is weighted based on its temporal proximity to the current time t_u. In addition, the TF-IWF vector of the user's currently location s_u at time t_u is included to capture short-term contextual relevance.

This design allows the semantic matching to reflect both the user's long-term spatial familiarity and the immediate context of their current activity.

Spatial proximity. We model spatial relevance using a distance-decay function $f_{\text{dist}}(d) = \exp(-\lambda_d \cdot d(s_u, i))$, where $d(s_u, i)$ is the geographical distance between user and item, and λ_d controls the decay rate. This formulation encourages recommending closer items while still allowing semantically relevant distant items to appear.

Public familiarity. To account for socially shared knowledge, $f_{\text{pop}}(i)$ reflects how frequently item i's location has been visited by the general public. This component promotes widely recognized places and enhances collective cognitive relevance, thereby increasing the exposure of popular and trustworthy content in recommendations.

By integrating personalized semantics, spatial constraints, and collective familiarity, our model provides passive recommendations that are both user-centric and socially grounded. This allows the system to suggest nearby and timely content that reflects both individual interests and shared patterns in the local community.

4 Experiments and Results

4.1 Data Description

To facilitate a robust empirical evaluation of *AskNearby* and its comparison with other LLM-based applications, we constructed a dataset of real-world, location-based user-generated content. This data was sourced from RedNote, one of China's foremost social media platforms for local discovery and lifestyle sharing. The study area is set in Shenzhen, a large metropolitan city in China. We programmatically collected geotagged posts associated with major landmarks (e.g., Nantou Ancient Town, Wutong Mountain) by targeting location-specific keywords.

To mitigate potential biases from RedNote's native recommendation algorithm, we implemented a three-stage data collection and processing methodology. The process began with a broad keyword search across the study area to gather a comprehensive and unfiltered corpus of candidate posts. This initial corpus was then subjected to a rigorous cleaning phase, which included deduplication, the removal of advertisements and spam, and sample-based verification of geographic information accuracy. Finally, from this refined corpus, we performed random sampling to constitute our final dataset, thereby ensuring it is a representative, rather than a platform-biased, sample of user-generated content.

The resulting dataset comprises approximately 20,000 posts and encompasses a diverse range of popular activity areas, including

信息项 i 的评分为：

$$\Psi(s_u, t_u, p_u, i) = f_{\text{sem}}(s_u, t_u, p_u, i)^{\alpha} \cdot f_{\text{dist}}(d(s_u, i))^{\beta} \cdot f_{\text{pop}}(i)^{\gamma}, \quad (6)$$

其中 α, β, γ 控制语义相关性、空间接近性和公众熟悉度的相对贡献。

在这一公式中，总体推荐得分被计算为三个部分的乘积：语义相关项 $f_{\text{sem}}(s_u, t_u, p_u, i)$，用于衡量用户认知特征与物品位置的时间功能语义之间的相似性；空间接近项 $f_{\text{dist}}(d(s_u, i))$，通过基于用户与物品空间距离的距离衰减函数来模拟地理相关性；以及公众熟悉度项 $f_{\text{pop}}(i)$，反映位置的整体受欢迎程度或访问频率，体现其集体认知显著性。

为了反映推荐模型中不同组件的相对重要性，我们为每个因素分配了指数 α、β 和 γ，它们作为乘法评分函数中的软权重使用。我们将这三个权重初始化为相等的 1.0，以避免施加任何先验偏好，使语义相关性、空间接近度和公众熟悉度能够同等地贡献于最终排名。我们承认，通过在验证集上进行网格搜索，可能可以找到更优的权重组合，这将留作未来的工作。

图2说明了这一评分框架背后的概念模型，展示了语义、空间和受欢迎度信号如何相互作用，以确定在不同情境下向用户推荐哪些内容。

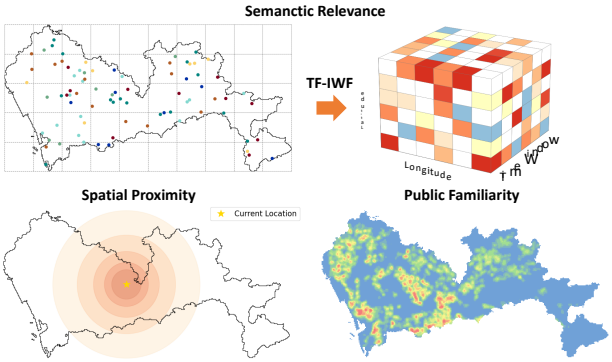

图2：AskNearby推荐框架示意图，结合语义相关性、空间接近性和公众熟悉度来计算个性化评分。

语义相关性。为了模拟用户如何认知地感知并将意义与地点关联，我们使用基于 TF-IWF（词频-逆文频）表示的方法实现语义组件[6]，该方法捕捉每个地点随时间变化的功能语义。该方案识别在局部显著但在不同时间段全球独特的地点属性。形式上，TF-IWF 分数计算如下：

$$\text{TF-IWF}_{k,g} = \frac{A_{k,g}}{\sum_i A_{i,g}} \cdot \log\left(\frac{\sum_h A_{k,h}}{\sum_i \sum_h A_{i,h}}\right), \quad (7)$$

其中 $A_{k,g}$ 表示在特定时间窗口内，位置 g 的功能属性 k（的数量，例如餐厅、办公室、公园。例如，一个位置在白天可能表现出与办公室相关的语义，而在晚上则转向餐饮。

语义相似度 $f_{\text{sem}}(s_u, t_u, p_u, i)$ 是通过将条目在时间 t_u 的位置的 TF-IWF 向量与用户的认知画像进行内积计算得出的。该画像是通过聚合 p_u 中的位置的 TF-IWF 向量构建的，p_u 是用户 u 经常访问的地点集合。每个位置的贡献根据其与当前时间 t_u 的时间接近性进行加权。此外，还包括用户在时间 t_u 的当前位置 s_u 的 TF-IWF 向量，以捕捉短期的上下文相关性。

该设计允许语义匹配反映用户的长期空间熟悉度以及他们当前活动的即时上下文。

空间接近性。我们使用距离衰减函数 $f_{\text{dist}}(d) = \exp(-\lambda_d \cdot d(s_u, i))$ 来建模空间相关性，其中 $d(s_u, i)$ 是用户与物品之间的地理距离，λ_d 控制衰减速率。该公式鼓励推荐更接近的物品，同时仍允许出现语义相关的远距离物品。

公众熟悉度。为了考虑社会共享的知识，$f_{\text{pop}}(i)$反映了项目i的位置被公众访问的频率。这个组成部分促进了广为人知的地点，并增强了集体认知相关性，从而增加了推荐中受欢迎和可信内容的曝光度。

通过整合个性化语义、空间约束和集体熟悉度，我们的模型提供既以用户为中心又具有社会基础的被动推荐。这使系统能够推荐附近及时的内容，同时反映个人兴趣和当地社区的共享模式。

4 实验与结果

4.1 数据描述

为了便于对 AskNearby 及其与其他基于大语言模型（LLM）应用的比较进行稳健的实证评估，我们构建了一个真实世界的基于位置的用户生成内容数据集。该数据来自 RedNote，这是中国领先的本地发现和生活方式分享社交媒体平台之一。研究区域设在中国的大都市深圳。我们通过面向特定位置的关键词，程序化地收集了与主要地标（如南头古镇、梧桐山）相关的带有地理标签的帖子。

为了减轻 RedNote 原生推荐算法可能带来的偏见，我们实施了三阶段的数据收集和处理方法。该过程首先在研究区域内进行广泛的关键词搜索，以收集全面且未筛选的候选帖子语料库。随后，对这一初始语料库进行了严格的清理阶段，包括去重、删除广告和垃圾信息，以及基于样本的地理信息准确性验证。最后，从这一精炼过的语料库中，我们进行了随机抽样，构成了最终的数据集，从而确保其是用户生成内容的代表性样本，而非受平台偏见影响的样本。

生成的数据集大约包括20,000条帖子，并涵盖各种受欢迎的活动领域，包括

historical sites like Nantou Ancient Town and natural attractions such as Wutong Mountain. This selection ensures comprehensive representation across a spectrum of urban functions, from commercial and recreational zones to key transportation hubs. Each record contains essential metadata such as post content, timestamp, user attributes (e.g., type), interaction metrics, semantic tags, and geographic coordinates. Table 1 provides a summary of the data schema.

Table 1: Key fields in the RedNote dataset

Field	Description
Title, Content	Textual description of the post
Timestamp	Posting date and time
Author, Type	Username and whether verified or not
Likes, Comments	Interaction counts
Semantic Tags	Keywords, topics, sentiment labels
Location Name	Place name extracted or tagged
Longitude, Latitude	Geographic coordinates (WGS84)
Media Type	Post format (text, image, video)
Original or Repost	Indicates whether the post is original content

This dataset serves as a rich source of localized information for benchmarking how *AskNearby* retrieves and recommends spatiotemporally relevant content compared to baseline LLM systems that lack grounded urban context. Furthermore, it enables fine-grained evaluation of user intent understanding, place-aware ranking, and recommendation diversity. The spatial distribution of the data is visualized in Figure 3, where the heatmap highlights the concentration of activity around major commercial and cultural hubs, illustrating the high spatial resolution and urban relevance of the dataset.

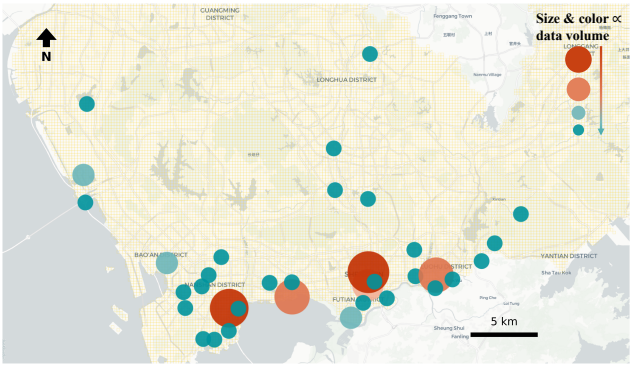

Figure 3: Spatial distribution of RedNote posts in Shenzhen.

4.2 Experimental Setup

Based on the dataset, we designed a set of local-life-oriented queries to assess the system's retrieval and generation performance.Notably, both the system and the dataset were originally in Chinese. For this paper, we provide a translated version in English to illustrate our approach. To ensure the linguistic alignment, we adopt ChatGLM-4 [30], one of the most advanced Chinese large language models available.

4.3 Evaluation Metrics

We use metrics covering two aspects to evaluate the retrieval and recommendation, respectively.

For retrieval ability, we use **Precision**, **Normalized Discounted Cumulative Gain (NDCG)**, **Hallucination Rate (HR)**, and **Spatial-Temporal Relevance (STR)** to evaluate.

- **Precision**: measures the relevance of retrieved posts. High precision indicates fewer irrelevant results. Since users usually only focus on the first few results, we adopt Precision@4 in this study.
- **Normalized Discounted Cumulative Gain (NDCG)**: evaluates ranking quality by incorporating graded relevance scores for each post (2 = highly relevant, 1 = moderately relevant, 0 = irrelevant). The metric computes a weighted cumulative gain for the top K results (here we use k=4) and normalizes it against the ideal ranking.
  Specifically,

$$\text{NDCG@4} = \frac{\text{DCG@4}}{\text{IDCG@4}} \tag{8}$$

$$\text{DCG@4} = \sum_{i=1}^{4} \frac{2^{\text{rel}_i} - 1}{\log_2(i+1)} \tag{9}$$

$$\text{IDCG@4} = \sum_{i=1}^{4} \frac{2^{\text{rel}_i^\star} - 1}{\log_2(i+1)} \tag{10}$$

  Here, $\text{rel}_i^\star$ represents the relevance scores in the ideal descending order.
- **Hallucination Rate (HR)**: proportion of responses generated by the model with non-existent or incorrectly attributed local information (e.g., fictitious attractions or incorrect prices).
- **Spatial-Temporal Relevance (STR)**: quantifies whether retrieved results are geographically constrained within the queried "proximity range" (e.g., a 1km living radius) and satisfy implicit temporal constraints (e.g., operational hours). Each result is scored 1 if both conditions are met; otherwise, it is 0.
- **Answer Quality (AQ)**: the overall integrity, coherence, and fluency of generated answers.
- **Match Score (MS)**: the degree of alignment between the answer and the query's intent, reflecting how effectively the response resolves the user's underlying need.

Furthermore, we introduce two LLM-evaluated metrics to complement several aspects that are hard to quantify: : (1) **Answer Quality (AQ)**: the overall integrity, coherence, and fluency of generated answers. (2) **Match Score (MS)**: the degree of alignment between the answer and the query's intent, reflecting how effectively the response resolves the user's underlying need. The prompts of LLMs for measuring **AQ** and **MS** are detailed in Appendix B.

For recommendation ability, we utilized **Hit Rate (Hit@K)**, **Normalized Discounted Cumulative Gain (NDCG@K)** and **Mean Reciprocal Rank (MRR)** to measure.

像南投古镇这样的历史遗址以及乌桐山等自然景点。这个选择确保在各类城市功能中都有全面的代表性，从商业和娱乐区到重要的交通枢纽。每条记录都包含必要的元数据，如帖子内容、时间戳、用户属性（例如类型）、交互指标、语义标签和地理坐标。表1提供了数据模式的摘要。

表 1：RedNote 数据集中的关键字段

Field	Description
Title, Content	Textual description of the post
Timestamp	Posting date and time
Author, Type	Username and whether verified or not
Likes, Comments	Interaction counts
Semantic Tags	Keywords, topics, sentiment labels
Location Name	Place name extracted or tagged
Longitude, Latitude	Geographic coordinates (WGS84)
Media Type	Post format (text, image, video)
Original or Repost	Indicates whether the post is original content

该数据集作为丰富的本地化信息来源，用于基准测试 AskNearby 在检索和推荐时空相关内容方面的表现，相较于缺乏扎实城市背景的基础 LLM 系统。此外，它还支持对用户意图理解、地点感知排序和推荐多样性的精细评估。数据的空间分布在图 3 中可视化，其中热力图突出显示了主要商业和文化枢纽周围的活动集中情况，展示了数据集的高空间分辨率和城市相关性。

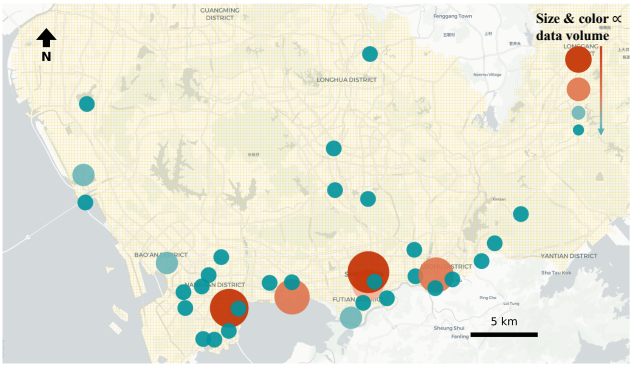

图3：深圳RedNote帖子的空间分布。

4.2 实验设置

基于该数据集，我们设计了一组面向本地生活的查询，以评估系统的检索和生成性能。值得注意的是，系统和数据集最初都是中文的。为了本文的目的，我们提供了英文翻译版本以说明我们的情况方法。为了确保语言一致性，我们采用了 ChatGLM-4 [30]，这是可用的最先进的中文大型语言模型之一。

4.3 评估指标

我们使用涵盖两个方面的指标分别评估检索和推荐。

对于检索能力，我们使用精确度、归一化折扣累积增益 (NDCG)、幻觉率 (HR) 和时空相关性 (STR) 进行评估。

- 精确率：衡量检索到的帖子相关性的指标。高精确率表示无关结果较少。由于用户通常只关注前几条结果，本研究采用 Precision@4。
- 归一化折扣累积增益（NDCG）：通过为每条帖子引入分级相关性评分来评估排序质量（2 = 高度相关，1 = 中度相关，0 = 不相关）。该指标计算前 K 个结果的加权累积增益（这里我们使用 k=4），并将其与理想排序进行归一化。

  具体来说，

$$\text{NDCG@4} = \frac{\text{DCG@4}}{\text{IDCG@4}} \tag{8}$$

$$\text{DCG@4} = \sum_{i=1}^{4} \frac{2^{\text{rel}_i} - 1}{\log_2(i+1)} \tag{9}$$

$$\text{IDCG@4} = \sum_{i=1}^{4} \frac{2^{\text{rel}_i^\star} - 1}{\log_2(i+1)} \tag{10}$$

  这里，$\text{rel}_i^\star$ 表示理想去冗余中的相关性得分升序。
- 幻觉率 (HR)：模型生成的包含不存在或错误归因的本地信息的回答的比例（例如，虚构的景点或错误的价格）。
- 时空相关性（STR）：量化检索结果是否在查询的“接近范围”（例如，1公里生活半径）内受到地理限制，并满足隐性的时间约束（例如，营业时间）。如果两个条件都满足，每个结果得分为1；否则得分为0。
- 答案质量（AQ）：生成答案的整体完整性、连贯性和流畅性。
- 匹配得分 (MS)：答案与查询意图之间的一致程度，反映了响应在多大程度上有效解决了用户的潜在需求。

此外，我们引入了两种由大语言模型（LLM）评估的指标，以补充一些难以量化的方面：(1) 回答质量（AQ）：生成回答的整体完整性、连贯性和流畅性。(2) 匹配得分（MS）：回答与查询意图的一致程度，反映回答解决用户潜在需求的有效性。用于测量 AQ 和 MS 的 LLM 提示详见附录 B。

对于推荐能力，我们使用命中率（Hit@K）、归一化折扣累积增益（NDCG@K）和平均倒数排名（MRR）进行测量。

- **Hit Rate (Hit@K)**: measures whether any ground-truth item appears within the top K recommended items. It reflects the coverage of relevant results, regardless of position. We report Hit@5 and Hit@10 in the experiment.
- **Normalized Discounted Cumulative Gain (NDCG@K)**: evaluates ranking quality by rewarding relevant items appearing earlier in the list. It is computed as described in the retrieval metrics, and we report NDCG@5 and NDCG@10.
- **Mean Reciprocal Rank (MRR)**: the reciprocal of the rank at which the first relevant item appears in the recommendation list. A higher MRR indicates that relevant content is shown earlier, enhancing user satisfaction.

These metrics effectively reflect user satisfaction in life-circle scenarios by measuring whether relevant posts appear early within a limited number of local recommendations.

Considering the nuanced judgment required by the evaluation, we employed a human evaluation methodology for the rigorous and objective comparison of our model against the baselines. The process involved three trained evaluators who assessed the outputs from all models, presented in a randomized and anonymized order to prevent bias. For relevance-based metrics like Precision@4 and NDCG@4, evaluators assigned relevance scores to each retrieved item, which were then used to compute the final values. For qualitative metrics such as Hallucination Rate (HR) and Answer Quality (AQ), they assigned scores directly based on predefined criteria including factual accuracy, spatiotemporal adherence, and contextual alignment. To ensure reliability, the final score for each metric was the average from the three evaluators, with inter-annotator agreement calculated and any significant discrepancies resolved through discussion.

4.4 Ablation Study

We conducted an ablation study to systematically evaluate the retrieval and recommendation capabilities of our system, respectively. This approach was chosen because our system is specifically engineered to address the novel LLIA problem by integrating dynamic user context, cognitive maps, and spatial-semantic features. Standard recommenders intrinsically lack these capabilities, and adapting them for this task would require non-trivial modifications, creating a significant risk of comparison bias. Therefore, an ablation study provides a more direct and insightful evaluation of each component in both retrieval and recommendation modules.

Results are summarized in Table 2 and Table 3. For retrieval, the full three-layer RAG (GraphRAG + VectorRAG + GeoRAG) achieves the best performance. Removing GraphRAG moderately reduces Precision@4 and NDCG@4, while excluding GeoRAG leads to the largest performance drop, with STR falling from 83.8% to 58.7% and hallucination rate rising, underscoring its role in grounding responses. VectorRAG alone performs worst across all metrics, showing that semantic similarity retrieval is insufficient without geographic and relational constraints. Through the experiments, we have validated the necessity of combining all three retrieval strategies to ensure both contextual relevance and geographic correctness in real-world use.

For recommendation, combining spatial proximity, public familiarity, and semantic relevance yields the best results. Dropping semantic or public cues degrades ranking quality and early-hit accuracy, while relying on spatial proximity alone performs worst, confirming that spatial context must be complemented with social and semantic signals.

4.5 Comparison Experiment

We then focus on evaluating retrieval performance. Our system is compared against several representative baseline models, including both LLM-based architectures. In addition to state-of-the-art general-purpose LLMs such as GPT-4o[4], we also include Chinese-developed models like DeepSeek[5] and Qwen[6]. Since these models do not inherently possess localized knowledge, we construct an external knowledge base for each and enable retrieval to ensure a fair comparison.

To reflect real-world usability, we further compare *AskNearby* with domain-specific or online map platforms, including RedNote[7], as well as the most widely adopted local map applications in China, such as Gaode Maps[8] and Baidu Maps[9]. These platforms have recently integrated advanced LLMs (e.g., DeepSeek-R1) to support natural language-based local information search and summarization. We conduct manual searches through the platform interface and collect the corresponding returned results.

Here are the introductions of our baseline models:

- **GPT-4o**: State-of-the-art multilingual LLMs with strong general reasoning and dialogue capabilities.
- **DeepSeek R1** and **Qwen3 Turbo**: High-performance Chinese LLMs optimized for local language understanding. These models demonstrate strong reasoning capabilities and complex Chinese instruction-following behavior, with significantly improved human preference alignment compared to previous generations.
- **RedNote**: A local lifestyle and information-sharing platform with user-generated content and keyword-based search.
- **Gaode Maps, Baidu Maps**: Leading map applications in China that offer location-based search with LLM-enhanced natural language interfaces.

Table 4 shows the evaluation results of our system. *AskNearby* outperforms most baselines across retrieval metrics. However, in precision, it ranks second (75.6%) slightly below RedNote (78.3%). This is likely attributed to the dominance of simple keyword matching tasks in the experimental queries (e.g., "Starbucks"), where RedNote's keyword-based engine excels. In contrast, RedNote struggles with queries requiring complex intent or spatiotemporal constraints (e.g., "a quiet coffee shop for afternoon work") while *AskNearby* excels with its superior semantic understanding capabilities, evident from its much higher NDCG@4 score (0.96 vs. 0.88). Notably, *AskNearby* far surpasses all baselines in STR, underscoring its ability to semantically and spatiotemporally retrieve relevant results—critical in real-world local information access. In addition, *AskNearby* excels in generating high-quality responses with minimal hallucinations. It effectively retrieves and summarizes relevant knowledge

[4] https://chatgpt.com/
[5] https://www.deepseek.com/
[6] https://chat.qwen.ai/
[7] https://www.xiaohongshu.com/
[8] https://gaode.com/
[9] https://map.baidu.com/

- 命中率 (Hit@K)：衡量是否有任何真实项出现在前 K 个推荐项中。它反映了相关结果的覆盖情况，而不考虑位置。在实验中，我们报告了 Hit@5 和 Hit@10。
- 归一化折扣累积增益 (NDCG@K)：通过奖励出现在列表前面的相关项目来评估排序质量。其计算方法如检索指标中所述，我们报告 NDCG@5 和 NDCG@10。
- 平均倒数排名（MRR）：推荐列表中第一个相关项目出现的排名的倒数。MRR 越高，表示相关内容越早显示，从而提升用户满意度。

这些指标通过衡量相关帖子是否在有限数量的本地推荐中提前出现，有效地反映了用户在生活圈场景中的满意度。

考虑到评估所需的细致判断，我们采用了人工评估方法，对我们的模型与基线模型进行严格且客观的比较。该过程涉及三名经过培训的评估员，他们评估所有模型的输出，这些输出以随机和匿名的顺序呈现，以防止偏见。对于基于相关性的指标，如 Precision@4 和 NDCG@4，评估员为每个检索项分配相关性分数，然后用这些分数计算最终值。对于定性指标，如虚构率（HR）和答案质量（AQ），评估员根据预定义的标准直接评分，这些标准包括事实准确性、时空一致性和语境匹配。为了确保可靠性，每个指标的最终得分为三位评估员的平均值，同时计算评估员间的一致性，并通过讨论解决任何显著分歧。

4.4 消融研究

我们进行了消融研究，以系统地评估我们系统的检索和推荐能力。选择这种方法是因为我们的系统专门设计用于通过整合动态用户上下文、认知地图和空间-语义特征来解决新颖的 LLIA 问题。标准推荐系统本质上缺乏这些能力，并且将其改造用于此任务需要非平凡的修改，从而产生显著的比较偏差风险。因此，消融研究能够对检索和推荐模块中的每个组件提供更直接且具有洞察力的评估。

结果总结在表2和表3中。对于检索任务，全三层RAG（GraphRAG +、VectorRAG +、GeoRAG）表现最佳。去掉GraphRAG会适度降低Precision@4和NDCG@4，而排除GeoRAG会导致性能下降最大，STR从83.8%下降到58.7%，幻觉率上升，凸显了其在回答落地中的作用。单独使用VectorRAG在所有指标上表现最差，表明没有地理和关系约束的语义相似性检索是不够的。通过实验，我们验证了结合三种检索策略的必要性，以确保在实际使用中既有上下文相关性又有地理准确性。

对于推荐，将空间接近性、公众熟悉度和语义相关性结合起来能产生最佳效果。放弃语义或公共提示会降低排序质量和早期命中准确率，而仅依赖空间接近性表现最差，这证实了空间上下文必须与社交和语义信号相辅相成。

4.5 比较实验

然后我们重点评估检索性能。我们的系统与几个具有代表性的基线模型进行了比较，包括基于大型语言模型（LLM）的架构。除了像 GPT-4o[4] 这样的最先进通用 LLM，我们还包括中国开发的模型，如 DeepSeek[5] 和 Qwen[6]。由于这些模型本身不具备本地化知识，我们为每个模型构建了一个外部知识库，并启用检索以确保公平比较。

为了反映现实世界的可用性，我们进一步将 AskNearby 与特定领域或在线地图平台进行比较，包括 RedNote[7]，以及在中国最广泛使用的本地地图应用，如高德地图[8] 和百度地图[9]。这些平台最近集成了先进的大型语言模型（例如 DeepSeek-R1），以支持基于自然语言的本地信息搜索和摘要。我们通过平台界面进行手动搜索，并收集相应返回的结果。

以下是我们基线模型的介绍：

- GPT-4o：具有强大通用推理和对话能力的最先进多语言大语言模型。
- DeepSeek R1 和 Qwen3 Turbo：高性能中文大模型，针对本地语言理解进行了优化。这些模型展现出强大的推理能力和复杂的中文指令遵循行为，与前几代相比，其与人类偏好的一致性显著提升。
- RedNote：一个本地生活方式和信息分享平台，具有用户生成内容和基于关键词的搜索功能。
- 高德地图、百度地图：中国领先的地图应用，提供基于位置的搜索，并配备LLM增强的自然语言界面。

表4显示了我们系统的评估结果。AskNearby在大多数检索指标上优于大多数基线。然而，在精确度方面，它排名第二（75.6%），略低于RedNote（78.3%）。这可能归因于实验查询中以简单关键词匹配任务为主（例如“星巴克”），而RedNote的基于关键词的引擎在此类任务中表现出色。相比之下，RedNote在处理需要复杂意图或时空限制的查询（例如“适合下午工作的安静咖啡馆”）时表现不佳，而AskNearby凭借其卓越的语义理解能力表现优异，这从其远高于RedNote的NDCG@4评分（0.96对0.88）可见一斑。值得注意的是，AskNearby在STR方面远超所有基线，这凸显了其在语义和时空上检索相关结果的能力——这一点在现实世界的本地信息访问中至关重要。此外，AskNearby在生成高质量响应方面同样表现出色，且出现虚构信息的情况极少。它能够有效地检索并总结相关知识。

[4] https://chatgpt.com/
[5] https://www.deepseek.com/
[6] https://chat.qwen.ai/
[7] https://www.xiaohongshu.com/
[8] https://gaode.com/
[9] https://map.baidu.com/

Table 2: Ablation study on different RAG-module combinations in *AskNearby*. The first row is the full model. Subsequent rows show the effect of removing one or more modules.

Method	Precision@4 (%)	NDCG@4	HR (%)	STR (%)	AQ	MS
AskNearby (All RAGs)	**75.6**	**0.96**	**2.5**	**83.8**	**3.9**	**4.2**
Geo + Vector (— GraphRAG)	70.8	0.91	3.2	82.9	3.6	3.8
Graph + Vector (— GeoRAG)	66.1	0.83	4.5	58.7	3.4	3.2
VectorRAG only (— GeoRAG, — GraphRAG)	59.7	0.78	5.1	53.6	3.1	2.9

Table 3: Ablation results of the *AskNearby* recommendation module (S = spatial proximity, P = public familiarity, Sem = semantic relevance).

Method	Hit@5	Hit@10	NDCG@5	NDCG@10	MRR
S + P + Sem	0.612	0.708	0.482	0.534	0.421
S + Sem	0.587	0.682	0.455	0.506	0.386
S + P	0.573	0.669	0.443	0.493	0.397
S only	0.542	0.638	0.412	0.461	0.357

Table 4: Evaluation results of *AskNearby* vs. baseline models (Bold: best, underline: second-best)

Method	Precision@4 ↑ (%)	NDCG@4 ↑	HR ↓ (%)	STR ↑
GPT-4o	58.3	0.86	15.8	41.7
DeepSeek-R1	72.8	0.92	15.8	64.2
Qwen-3-turbo	70.8	0.92	15.0	62.5
RedNote	**78.3**	0.88	13.1	56.9
Gaode maps	50.0	0.65	9.3	41.0
Baidu maps	65.0	0.71	3.0	50.0
AskNearby (Ours)	75.6	**0.96**	**2.5**	**83.8**

grounded in local context, leading to informative and trustworthy answers.

4.6 Qualitative Analysis

To demonstrate the practical effectiveness of our system, we further conduct a qualitative analysis in real-world local information scenarios. Figure 4 presents a representative case in which a user poses the context-specific query: "Where can I find healthcare services inside the neighborhood?". We compare the response generated by *AskNearby* against a baseline LLM-based application.

As the figure illustrates, the baseline application, while identifying semantically related content, ultimately hallucinates a non-existent address. This failure stems from a lack of accurate, localized knowledge grounding, which can mislead the user. In stark contrast, *AskNearby* effectively synthesizes information from relevant posts to extract and present the correct service location, substantiated with supporting details such as contact information.

In addition to this example, generic LLMs, even when enhanced with retrieval capabilities, often suffer from hallucination when dealing with fine-grained, up-to-date local information. This further indicates the superiority of our AskNearby in satisfying the needs for the retrieval and recommendation of local information within the life circle.

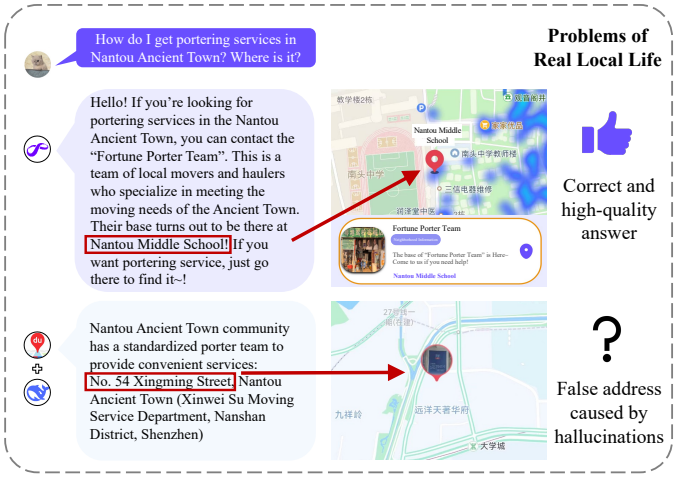

Figure 4: Answers to a local life problem from AskNearby and Baidu maps (powered by DeepSeek-R1).

4.7 Deployed System Performance

To validate the effectiveness and scalability of *AskNearby*, we conducted extensive field deployments in two diverse real-world environments: a university campus (PKU-SZ) and a high-density, mixed-use urban community (Nantou Ancient City). Our field deployments yielded significant positive outcomes in two diverse environments. On a university campus, *AskNearby* saw rapid adoption, achieving a 46% uptake among new students within 30 days. In a dense urban community (Nantou, 38,000 residents/km^2), the system fostered significant hyperlocal engagement, as evidenced by residents initiated 4.1 queries per month on average, enabling 1,472 offline interactions.

Also, we conduct manual evaluation by inviting 40 regular users (**User**) and 10 domain experts (**Expert**) in urban planning to assess *AskNearby* against baselines based on their individual queries.

表 2：AskNearby 中不同 RAG 模块组合的消融研究。第一行是完整模型。随后的各行显示移除一个或多个模块的效果。

Method	Precision@4 (%)	NDCG@4	HR (%)	STR (%)	AQ	MS
AskNearby (All RAGs)	**75.6**	**0.96**	**2.5**	**83.8**	**3.9**	**4.2**
Geo + Vector (— GraphRAG)	70.8	0.91	3.2	82.9	3.6	3.8
Graph + Vector (— GeoRAG)	66.1	0.83	4.5	58.7	3.4	3.2
VectorRAG only (— GeoRAG, — GraphRAG)	59.7	0.78	5.1	53.6	3.1	2.9

表 3：AskNearby 推荐模块的消融实验结果（S = 空间接近性，P = 公众熟悉度，Sem = 语义相关性）。

Method	Hit@5	Hit@10	NDCG@5	NDCG@10	MRR
S + P + Sem	0.612	0.708	0.482	0.534	0.421
S + Sem	0.587	0.682	0.455	0.506	0.386
S + P	0.573	0.669	0.443	0.493	0.397
S only	0.542	0.638	0.412	0.461	0.357

表4：AskNearby与基线模型的评估结果（加粗：最佳，下划线：第二佳）

Method	Precision@4 ↑ (%)	NDCG@4 ↑	HR ↓ (%)	STR ↑
GPT-4o	58.3	0.86	15.8	41.7
DeepSeek-R1	72.8	<u>0.92</u>	15.8	<u>64.2</u>
Qwen-3-turbo	70.8	<u>0.92</u>	15.0	62.5
RedNote	**78.3**	0.88	13.1	56.9
Gaode maps	50.0	0.65	9.3	41.0
Baidu maps	65.0	0.71	<u>3.0</u>	50.0
AskNearby (Ours)	<u>75.6</u>	**0.96**	**2.5**	**83.8**

根植于本地语境，从而提供有信息量且值得信赖的答案。

4.6 定性分析

为了展示我们系统的实际有效性，我们进一步在现实世界的本地信息场景中进行了定性分析。图4展示了一个代表性案例，其中用户提出了特定情境的查询：“我在哪里可以找到社区内的医疗服务？”。我们将AskNearby生成的回复与基于LLM的基线应用进行了比较。

如图所示，基线应用程序虽然能够识别语义相关的内容，但最终却虚构了一个不存在的地址。这一失败源于缺乏准确、本地化的知识基础，可能会误导用户。与此鲜明对比的是，AskNearby 能够有效地从相关帖子中综合信息，以提取并呈现正确的服务地点，并提供如联系方式等支持性细节。

除了这个例子之外，通用大语言模型（即使在增强了检索能力后）在处理精细、最新的本地信息时，仍然经常出现幻觉。这进一步说明了我们 AskNearby 在满足生活圈内本地信息检索和推荐需求方面的优势。

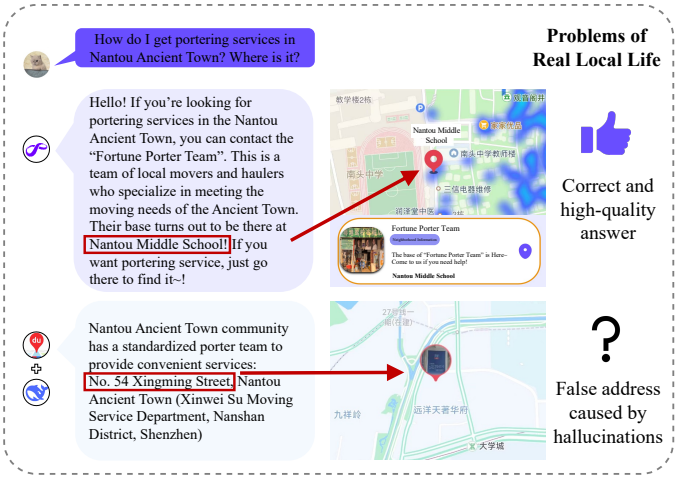

图4：来自 AskNearby 和百度地图（由 DeepSeek-R1 提供支持）对本地生活问题的回答。

4.7 部署系统性能

为了验证 AskNearby 的有效性和可扩展性，我们在两个不同的真实环境中进行了广泛的实地部署：一个大学校园（PKU-SZ）和一个高密度、混合用途的城市社区（南头古城）。我们的实地部署在这两种不同环境中都取得了显著的积极成果。在大学校园中，AskNearby 得到快速推广，在30天内新生的使用率达到了46%。在一个密集的城市社区（南头，38,000居民/km²）中，该系统促进了显著的本地互动，数据显示居民平均每月发起4.1次查询，实现了1,472次线下互动。

此外，我们通过邀请40名普通用户（User）和10名城市规划领域的专家（Expert），根据他们各自的查询，对AskNearby与基线进行评估，从而进行人工评估。

Table 5: Deployed system performance of *AskNearby* vs. baseline models (Bold: best, underline: second-best)

Method	AQ ↑	MS ↑	User ↑	Expert ↑
GPT-4o	3.5	3.8	3.9	3.1
DeepSeek-R1	3.7	4.0	4.0	4.3
Qwen-3-turbo	3.6	3.7	3.2	3.3
RedNote	2.2	3.1	2.7	2.5
Gaode maps	3.1	3.2	2.8	3.2
Baidu maps	3.8	3.9	3.7	3.8
AskNearby (Ours)	**3.9**	**4.2**	**4.6**	**4.5**

Performance was measured across four key dimensions: AQ and MS, alongside manual ratings from Users and Experts. The results, summarized in Table 5, demonstrate the superiority of our system, confirming its effectiveness in real-world scenarios.

This robust performance, validated in both large-scale deployments and controlled evaluations, underscores the real-world value of our approach, enhancing information accessibility and serving as a social connector that fosters neighborhood interaction.

5 Conclusion

In this paper, we introduce the problem of LLIA, which focuses on enabling residents to efficiently acquire relevant, timely, and context-aware information within their neighborhood. Unlike traditional search or recommendation systems that overlook fine-grained spatiotemporal dynamics and user-specific spatial cognition, LLIA highlights the importance of integrating geographic proximity, temporal constraints, and personalized perceptions in local information access. To address this challenge, we propose *AskNearby*, an AI-driven community platform integrating a three-layer retrieval-augmented generation framework with a cognitive map-based recommendation model.

Experiments on real-world datasets demonstrate that *AskNearby* significantly outperforms both LLM-based baselines and existing local service platforms in retrieval accuracy, contextual alignment, and hallucination mitigation. Qualitative studies and real-world deployments further validate its effectiveness, demonstrating both accurate localized knowledge grounding and the empowerment of residents with hyperlocal knowledge. By supporting both active search and passive recommendation at the neighborhood scale, as well as the outstanding ability of understanding natural language, *AskNearby* transforms static POIs into more local scenarios, allowing residents to know not only "what is there," but more importantly, "what it is like there," thereby effectively bridges the gap between physical proximity and true psychological and lifestyle accessibility. This brings the 15-minute city vision closer to reality. Moreover, unearthing fine-grained urban information overlooked by traditional maps, our work activates rich local resources, transforming communities from static geographical entities into dynamic socio-economic networks, greatly enriching the concept of the "15-minute city" and opening new possibilities for human-centric spatial computing in urban environments.

While our *AskNearby* system shows promising results, we acknowledge certain limitations and identify key directions for future work. We plan to advance our research along three main fronts. First, we aim to refine the core cognitive map model, focusing on more sophisticated, adaptive mechanisms to capture individual user nuances and their evolving preferences. Second, to enhance the scientific rigor and generalizability of our findings, we will significantly expand our data foundation and refine our evaluation methodologies. This involves diversifying data sources and geographical coverage beyond initial settings, alongside exploring more robust assessment protocols. Finally, we will focus on enhancing the practical utility and real-world applicability of *AskNearby*. This includes enriching its knowledge base with a wider array of information sources to improve overall accuracy and versatility, and conducting broader deployments to validate its effectiveness across diverse urban contexts and contribute to the development of intelligent localized services.

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References

[1] Timur Abbiasov, Cate Heine, Sadegh Sabouri, Arianna Salazar-Miranda, Paolo Santi, Edward Glaeser, and Carlo Ratti. 2024. The 15-minute city quantified using human mobility data. *Nature Human Behaviour* 8, 3 (2024), 445–455.

[2] Mohammad Aliannejadi and Fabio Crestani. 2018. Personalized context-aware point of interest recommendation. *ACM Transactions on Information Systems (TOIS)* 36, 4 (2018), 1–28.

[3] Mohammad Aliannejadi, Dimitrios Rafailidis, and Fabio Crestani. 2019. A joint two-phase time-sensitive regularized collaborative ranking model for point of interest recommendation. *IEEE Transactions on Knowledge and Data Engineering* 32, 6 (2019), 1050–1063.

[4] Matteo Bruno, Hygor Piaget Monteiro Melo, Bruno Campanelli, and Vittorio Loreto. 2024. A universal framework for inclusive 15-minute cities. *Nature Cities* 1, 10 (2024), 633–641.

[5] Shahaf Donio and Eran Toch. 2025. Neighborhood Disparities in Smart City Service Adoption. *arXiv preprint arXiv:2501.04363* (2025).

[6] Zhipeng Gui, Yunzeng Sun, Le Yang, Dehua Peng, Fa Li, Huayi Wu, Chi Guo, Wenfei Guo, and Jianya Gong. 2021. LSI-LSTM: An attention-aware LSTM for real-time driving destination prediction by considering location semantics and location importance of trajectory points. *Neurocomputing* 440 (2021), 72–88.

[7] Qiuhan Han, Atsushi Yoshikawa, and Masayuki Yamamura. 2025. Adapting Large Language Model for Spatio-Temporal Understanding in Next Point-of-Interest Prediction. In *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 1–5.

[8] Erum Haris, Anthony G Cohn, and John G Stell. 2024. Exploring spatial representations in the historical lake district texts with llm-based relation extraction. *arXiv preprint arXiv:2406.14336* (2024).

[9] Junlin He, Tong Nie, and Wei Ma. 2025. Geolocation representation from large language models are generic enhancers for spatio-temporal learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 17094–17104.

[10] Zongcai Huang, Peng Peng, Feng Lu, and He Zhang. 2025. An LLM-Based Method for Quality Information Extraction From Web Text for Crowed-Sensing Spatiotemporal Data. *Transactions in GIS* 29, 1 (2025).

[11] Amir Reza Khavarian-Garmsir, Ayyoob Sharifi, and Ali Sadeghi. 2023. The 15-minute city: Urban planning and design efforts toward creating sustainable neighborhoods. *Cities* 132 (2023), 104101.

[12] Boyang Li and Wenjia Zhang. 2024. Enhancing Pedestrian Route Choice Models Through Maximum-Entropy Deep Inverse Reinforcement Learning With Individual Covariates (MEDIRL-IC). *IEEE Transactions on Intelligent Transportation Systems* 25, 12 (2024), 20446–20463. doi:10.1109/TITS.2024.3457680

[13] Peibo Li, Maarten de Rijke, Hao Xue, Shuang Ao, Yang Song, and Flora D Salim. 2024. Large language models for next point-of-interest recommendation. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 1463–1472.

[14] Zhonghang Li, Lianghao Xia, Jiabin Tang, Yong Xu, Lei Shi, Long Xia, Dawei Yin, and Chao Huang. 2024. Urbangpt: Spatio-temporal large language models. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 5351–5362.

表 5：AskNearby 与基线模型的已部署系统性能（加粗：最佳，带下划线：第二佳）

Method	AQ ↑	MS ↑	User ↑	Expert ↑
GPT-4o	3.5	3.8	3.9	3.1
DeepSeek-R1	3.7	<u>4.0</u>	<u>4.0</u>	<u>4.3</u>
Qwen-3-turbo	3.6	3.7	3.2	3.3
RedNote	2.2	3.1	2.7	2.5
Gaode maps	3.1	3.2	2.8	3.2
Baidu maps	<u>3.8</u>	3.9	3.7	3.8
AskNearby (Ours)	**3.9**	**4.2**	**4.6**	**4.5**

性能通过四个关键维度进行衡量：AQ 和 MS，以及来自用户和专家的人工评分。结果总结在表 5 中，显示了我们系统的优势，确认了其在实际场景中的有效性。

这一强大性能在大规模部署和受控评估中均得到验证，凸显了我们方法的实际价值，增强了信息可获取性，并作为促进邻里互动的社会纽带。

5 结论

在本文中，我们介绍了LLIA问题，该问题侧重于使居民能够高效获取其社区内相关、及时且具有情境感知的信息。与忽略细粒度时空动态和用户特定空间认知的传统搜索或推荐系统不同，LLIA强调在本地信息获取中整合地理邻近性、时间约束和个性化感知的重要性。为了解决这一挑战，我们提出了AskNearby，一个由人工智能驱动的社区平台，结合了三层检索增强生成框架和基于认知地图的推荐模型。

在真实世界的数据集上的实验表明，AskNearby 在检索准确性、上下文对齐和幻觉缓解方面，显著优于基于大型语言模型的基线方法和现有的本地服务平台。定性研究和实际部署进一步验证了其有效性，展示了准确的本地化知识基础以及赋能居民掌握超本地化知识的能力。通过支持社区范围内的主动搜索和被动推荐，以及出色的自然语言理解能力，AskNearby 将静态POI转化为更本地化的场景，使居民不仅了解“那里有什么”，更重要的是了解“那里是什么样子的”，从而有效地弥合物理距离与真实心理和生活方式可达性之间的差距。这使“15分钟城市”的愿景更接近现实。此外，还能够挖掘传统地图忽略的细粒度城市信息。

虽然我们的AskNearby系统显示出有希望的结果，但我们承认存在某些局限性，并确定了未来工作的关键方向。我们计划沿着三个主要方向推进我们的研究。

首先，我们旨在完善核心认知地图模型，重点关注更复杂、适应性更强的机制，以捕捉个体用户的细微差异及其不断变化的偏好。其次，为了提高我们研究成果的科学严谨性和普适性，我们将大幅扩展数据基础，并改进评估方法。这包括在超出初始环境的情况下多样化数据来源和地理覆盖范围，同时探索更稳健的评估方案。最后，我们将专注于提升AskNearby的实际应用价值和现实适用性。这包括通过更广泛的信息来源丰富其知识库，以提高整体的准确性和多功能性，并进行更广泛的部署，以验证其在不同城市环境中的有效性，并为智能本地化服务的发展作出贡献。

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参考文献

[1] Timur Abbiasov、Cate Heine、Sadegh Sabouri、Arianna Salazar-Miranda、Paolo Santi、Edward Glaeser 和 Carlo Ratti. 2024. 使用人类移动数据量化的15分钟城市. 《自然人类行为》8卷, 第3期 (2024), 445–455. [2] Mohammad Aliannejadi 和 Fabio Crestani. 2018. 个性化的上下文感知兴趣点推荐. 《ACM信息系统学报》(TOIS) 36卷, 第4期 (2018), 1–28. [3] Mohammad Aliannejadi、Dimitrios Rafailidis 和 Fabio Crestani. 2019. 一个用于兴趣点推荐的联合两阶段时间敏感正则化协作排序模型. 《IEEE知识与数据工程汇刊》32卷, 第6期 (2019), 1050–1063. [4] Matteo Bruno、Hygor Piaget Monteiro Melo、Bruno Campanelli 和 Vittorio Loreto. 2024. 面向包容性15分钟城市的通用框架. 《自然城市》1卷, 第10期 (2024), 633–641. [5] Shahaf Donio 和 Eran Toch. 2025. 智慧城市服务采纳中的社区差异. arXiv 预印本 arXiv:2501.04363 (2025). [6] Zhipeng Gui

[15] Hui Luo, Jingbo Zhou, Zhifeng Bao, Shuangli Li, J. Shane Culpepper, Haochao Ying, Hao Liu, and Hui Xiong. 2020. Spatial Object Recommendation with Hints: When Spatial Granularity Matters. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval* (Virtual Event, China) *(SIGIR '20)*. Association for Computing Machinery, New York, NY, USA, 781–790. doi:10.1145/3397271.3401090

[16] Shunmei Meng, Huihui Wang, Qianmu Li, Yun Luo, Wanchun Dou, and Shaohua Wan. 2018. Spatial-Temporal Aware Intelligent Service Recommendation Method Based on Distributed Tensor factorization for Big Data Applications. *IEEE Access* 6 (2018), 59462–59474. doi:10.1109/ACCESS.2018.2872351

[17] Carlos Moreno, Zaheer Allam, Didier Chabaud, Catherine Gall, and Florent Pratlong. 2021. Introducing the "15-Minute City": Sustainability, resilience and place identity in future post-pandemic cities. *Smart cities* 4, 1 (2021), 93–111.

[18] Hang Ni, Fan Liu, Xinyu Ma, Lixin Su, Shuaiqiang Wang, Dawei Yin, Hui Xiong, and Hao Liu. 2025. TP-RAG: Benchmarking Retrieval-Augmented Large Language Model Agents for Spatiotemporal-Aware Travel Planning. *arXiv preprint arXiv:2504.08694* (2025).

[19] Lara Quijano-Sánchez, Iván Cantador, María E Cortés-Cediel, and Olga Gil. 2020. Recommender systems for smart cities. *Information systems* 92 (2020), 101545.

[20] Ting Shen, Haiquan Chen, and Wei-Shinn Ku. 2018. Time-aware location sequence recommendation for cold-start mobile users. In *Proceedings of the 26th ACM SIGSPATIAL international conference on advances in geographic information systems*. 484–487.

[21] Yuanyuan Tian, Wenwen Li, Lei Hu, Xiao Chen, Michael Brook, Michael Brubaker, Fan Zhang, and Anna K. Liljedahl. [n. d.]. Advancing Large Language Models for Spatiotemporal and Semantic Association Mining of Similar Environmental Events. ([n. d.]). doi:10.1111/tgis.13282

[22] Tianxing Wang and Can Wang. 2024. Embracing LLMs for Point-of-Interest Recommendations. *IEEE Intelligent Systems* 39, 1 (2024), 56–59. doi:10.1109/MIS.2023.3343489

[23] Shitao Xiao, Zheng Liu, Peitian Zhang, and Niklas Muennighoff. 2023. C-Pack: Packaged Resources To Advance General Chinese Embedding. arXiv:2309.07597 [cs.CL]

[24] Fengli Xu, Qi Wang, Esteban Moro, Lin Chen, Arianna Salazar Miranda, Marta C González, Michele Tizzoni, Chaoming Song, Carlo Ratti, Luis Bettencourt, et al. 2025. Using human mobility data to quantify experienced urban inequalities. *Nature Human Behaviour* (2025), 1–11.

[25] Wenping Yin, Yong Xue, Ziqi Liu, Hao Li, and Martin Werner. 2025. LLM-enhanced disaster geolocalization using implicit geoinformation from multimodal data: A case study of Hurricane Harvey. *International Journal of Applied Earth Observation and Geoinformation* 137 (2025), 104423.

[26] Phatpicha Yochum, Liang Chang, Tianlong Gu, and Manli Zhu. 2020. Linked Open Data in Location-Based Recommendation System on Tourism Domain: A Survey. *IEEE Access* 8 (2020), 16409–16439. doi:10.1109/ACCESS.2020.2967120

[27] Dazhou Yu, Riyang Bao, Gengchen Mai, and Liang Zhao. 2025. Spatial-RAG: Spatial Retrieval Augmented Generation for Real-World Spatial Reasoning Questions. *arXiv preprint arXiv:2502.18470* (2025).

[28] Xinli Yu, Zheng Chen, Yuan Ling, Shujing Dong, Zongyi Liu, and Yanbin Lu. 2023. Temporal data meets LLM–explainable financial time series forecasting. *arXiv preprint arXiv:2306.11025* (2023).

[29] Chenhan Yuan, Qianqian Xie, Jimin Huang, and Sophia Ananiadou. 2024. Back to the future: Towards explainable temporal reasoning with large language models. In *Proceedings of the ACM Web Conference 2024*. 1963–1974.

[30] Team GLM: Aohan Zeng, Bin Xu, Bowen Wang, Chenhui Zhang, Da Yin, Dan Zhang, Diego Rojas, Guanyu Feng, Hanlin Zhao, Hanyu Lai, Hao Yu, Hongning Wang, Jiadai Sun, Jiajie Zhang, Jiale Cheng, Jiayi Gui, Jie Tang, Jing Zhang, Jingyu Sun, Juanzi Li, Lei Zhao, Lindong Wu, Lucen Zhong, Mingdao Liu, Minlie Huang, Peng Zhang, Qinkai Zheng, Rui Lu, Shuaiqi Duan, Shudan Zhang, Shulin Cao, Shuxun Yang, Weng Lam Tam, Wenyi Zhao, Xiao Liu, Xiao Xia, Xiaohan Zhang, Xiaotao Gu, Xin Lv, Xinghan Liu, Xinyi Liu, Xinyue Yang, Xixuan Song, Xunkai Zhang, Yifan An, Yifan Xu, Yilin Niu, Yuantao Yang, Yueyan Li, Yushi Bai, Yuxiao Dong, Zehan Qi, Zhaoyu Wang, Zhen Yang, Zhengxiao Du, Zhenyu Hou, and Zihan Wang. 2024. ChatGLM: A Family of Large Language Models from GLM-130B to GLM-4 All Tools. arXiv:2406.12793 [cs.CL] https://arxiv.org/abs/2406.12793

[31] Pengpeng Zhao, Anjing Luo, Yanchi Liu, Jiajie Xu, Zhixu Li, Fuzhen Zhuang, Victor S. Sheng, and Xiaofang Zhou. 2022. Where to Go Next: A Spatio-Temporal Gated Network for Next POI Recommendation. *IEEE Transactions on Knowledge and Data Engineering* 34, 5 (2022), 2512–2524. doi:10.1109/TKDE.2020.3007194

[32] Zihuai Zhao, Wenqi Fan, Jiatong Li, Yunqing Liu, Xiaowei Mei, Yiqi Wang, Zhen Wen, Fei Wang, Xiangyu Zhao, Jiliang Tang, and Qing Li. 2024. Recommender Systems in the Era of Large Language Models (LLMs). *IEEE Transactions on Knowledge and Data Engineering* 36, 11 (2024), 6889–6907. doi:10.1109/TKDE.2024.3392335

[33] Yu Zheng, Licia Capra, Ouri Wolfson, and Hai Yang. 2014. Urban computing: concepts, methodologies, and applications. *ACM Transactions on Intelligent Systems and Technology (TIST)* 5, 3 (2014), 1–55.

[34] Yu Zheng and Xing Xie. 2011. Learning travel recommendations from user-generated GPS traces. *ACM Transactions on Intelligent Systems and Technology (TIST)* 2, 1 (2011), 1–29.

A Demonstration of Our AskNearby System

Figure 5 presents the screenshots of our developed AskNearby system, which is designed to enhance local life information accessibility through both recommendation and retrieval functionalities.

Subfigure (a) illustrates the **Recommendation Page**, where users receive personalized local content suggestions based on location, time, and cognitive preferences. The left panel features a vertically scrollable list of recommended posts concerning nearby locations or events, detailed with text descriptions, timestamps, spatial distances, and multimedia content. Adjacent to this list, an interactive map interface is provided in the right panel. On this map, users can adjust the location pointer to refresh their current view and receive recommendations for different areas.

Subfigure (b) shows the **Retrieval Page**, which supports natural-language-based active querying. Users can input open-domain questions about their surroundings, such as "Where are the toilets nearby?", and the system retrieves relevant spatial entities from the spatiotemporal knowledge base. The retrieval results are displayed in textual format, providing detailed descriptions, and visually on the map, allowing users to explore spatial relations interactively. The system also suggests example queries to facilitate information discovery.

B LLM Evaluation Prompts

B.1 LLM Evaluation Prompt for Local Information Retrieval

To assess AQ and MS metrics, we utilized Gemini 2.5 as the evaluator, leveraging its established high correlation with human judgment. The selection of an external model, distinct from our internal ChatGLM-4 and other comparative models, was a deliberate measure to prevent self-evaluation bias. Here are the prompts.

```
You are serving as a rigorous reviewer and need to score
     the following response. Please provide two integer
     scores on a 1-5 scale (5 being the best):

# Criteria
Answer Quality (evaluating the overall integrity,
     coherence, and fluency of generated answers)
Match Score (assessing the degree of alignment between the
     answer and the query's intent, reflecting how
     effectively the response addresses the user's
     underlying need)

# Output Format (only return JSON)
{
  "Answer Quality": <1-5>,
  "Match Score": <1-5>,
  "Comment": "<concise justification in 50 characters or
     less>"
}

# Input
## Question
(User's input query)
```

[15] 罗辉、周景博、鲍志峰、李双利、J. Shane Culpepper、应浩超、刘浩、熊辉。2020。带提示的空间对象推荐：当空间粒度重要时。载于《第43届国际ACM SIGIR信息检索研究与开发会议论文集》（虚拟会议，中国）（SIGIR '20）。美国纽约：计算机协会，781–790。doi:10.1145/3397271.3401090[16] 孟顺梅、王慧慧、李千木、罗芸、窦万春、万绍华。2018。基于分布式张量分解的大数据应用时空感知智能服务推荐方法。IEEE Access 6 (2018), 59462–59474. doi:10.1109/ACCESS.2018.2872351[17] Carlos Moreno、Zaheer Allam、Didier Chabaud、Catherine Gall、Florent Pratlong。2021。引入“15分钟城市”：未来后疫情城市的可持续性、韧性与地方身份。智能城市 4, 1 (2021), 93–111.[18] 倪航、刘凡、马新宇、L

[34] 余正，谢行。2011。基于用户生成的GPS轨迹学习旅行推荐。《ACM智能系统与技术汇刊（TIST）》2卷，1期（2011），第1–29页。

我们的 AskNearby 系统演示

图5展示了我们开发的AskNearby系统的截图，该系统旨在通过推荐和检索功能提升本地生活信息的可访问性。

子图 (a) 展示了推荐页面，用户可以根据位置、时间和认知偏好接收个性化的本地内容推荐。左侧面板呈现了一个可垂直滚动的推荐帖子列表，内容涉及附近的地点或活动，并配有文字描述、时间戳、空间距离及多媒体内容。在该列表旁边，右侧面板提供了一个交互式地图界面。在地图上，用户可以调整位置指针以刷新当前视图，并获取不同区域的推荐内容。

子图(b)显示了检索页面，该页面支持基于自然语言的主动查询。用户可以输入关于周围环境的开放域问题，例如“附近的厕所在哪里？”，系统会从时空知识库中检索相关的空间实体。检索结果以文本格式显示，提供详细描述，并在地图上以可视化方式呈现，允许用户互动式地探索空间关系。系统还会提供示例查询以便于信息发现。

B LLM 评估提示 B.1 用于本地信息检索的 LLM 评估提示

为了评估 AQ 和 MS 指标，我们使用 Gemini 2.5 作为评估器，利用其与人工评判的高度相关性。选择一个外部模型，与我们内部的 ChatGLM-4 及其他对比模型不同，是为了防止自我评估偏差。这些是提示语。

```
你正在担任严格的审稿人，需要对以下回应进行评分。请提供两个整数分数，范围为1-5（5分为最佳）：


# 标准
回答质量（评估生成答案的整体完整性、一致性和流畅性）匹配得分（评估答案与查询意图的一致程度，反映响应在多大程度上有效满足用户的潜在需求）




# 输出格式（仅返回 JSON） { "答案质量": <1-5>, "匹配得分": <1-5>, "评论": "<简明理由，不超过50个字符>" }# 输入 ## 问题（用户输入查询）
```

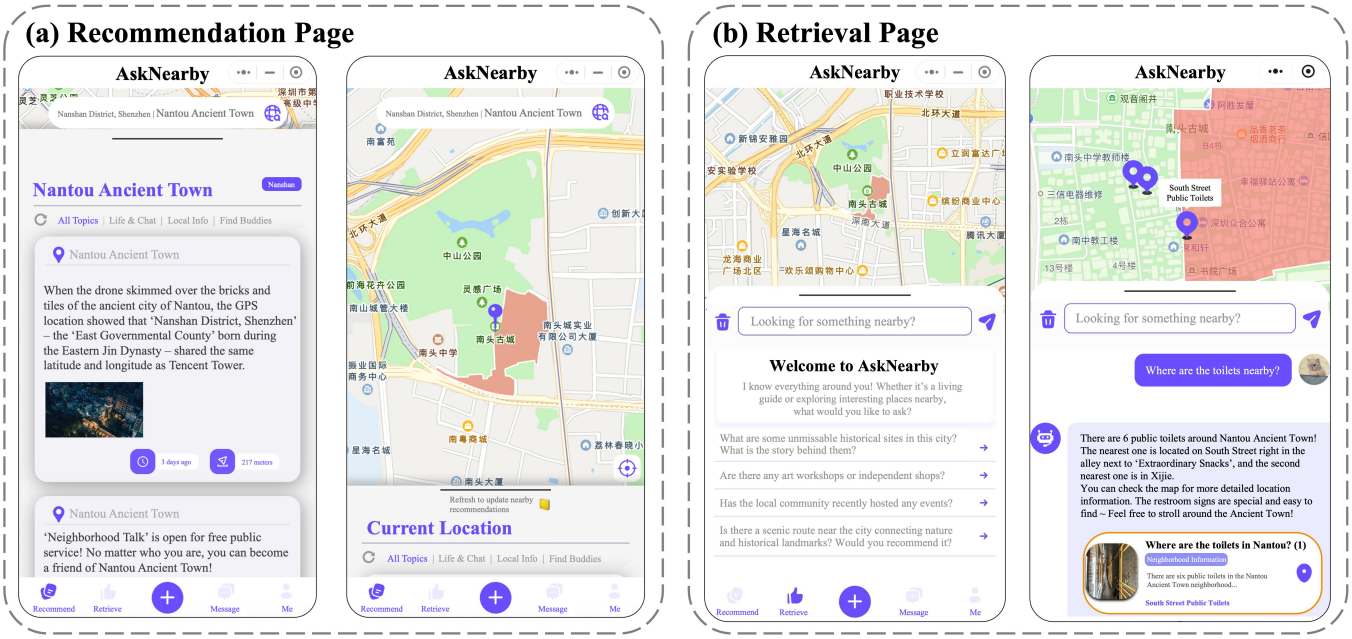

Figure 5: Screenshots of the AskNearby system: (a) Recommendation page and (b) Retrieval page.

```
## Response
(System-generated answer)
```

B.2 LLM Evaluation Prompt for Community Life-Circle Recommendations

```
You are an evaluator for a life-circle recommendation
    system.
Your current time is {Time}, and your location is
    {Location}.
Based on the recommended post list (see below), output the
    following evaluation as JSON:

1. How many posts are relevant to your life circle (within
    5 km or a familiar neighborhood)?
2. For each post, give a match score (0.0 to 1.0)
    representing how useful the content is to you.
3. At which position (starting from 1) is the post that you
    find most interesting?

Output format (only return JSON):
{
  "relevant_post_count": <int>,
  "match_score": [<float>, <float>, ...],
  "top_interest_position": <int>
}

Post Info:
[
  {
    "sname": "Shenzhen University Town",
    "title": "Flea Market | Second-hand Bicycle for Sale",
    "post_content": "Selling a nearly new bike on campus,
     80% new, sincere offer",
    "time": "2024-03-01 14:52:00",
    "latitude": 22.590,
    "longitude": 113.943,
    "tag": ["Shenzhen", "Nanshan", "Campus", "Second-hand"]
  },
  ...
]

Note: Only return valid JSON. Do not include explanations
    or extra text.
```

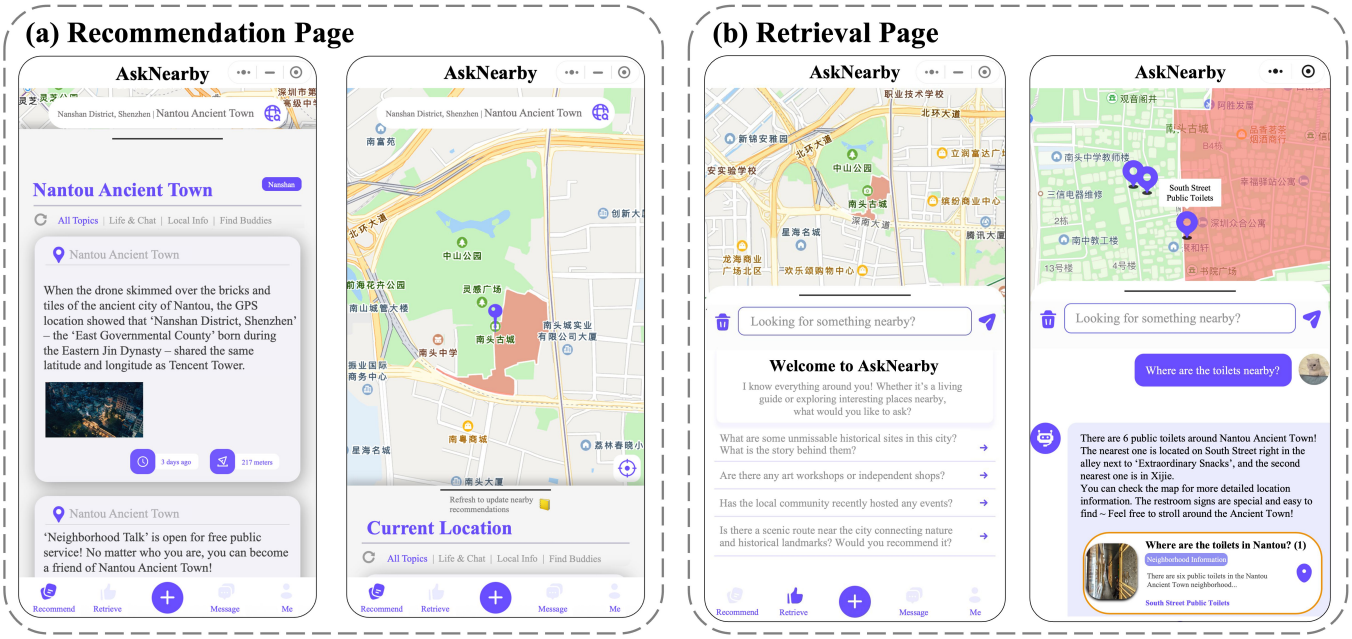

图5：AskNearby 系统的截图：（a）推荐页面和（b）检索页面。

回应（系统生成的答案）

B.2 社区生活圈推荐的大型语言模型评估提示

你是一个生活圈推荐系统的评估员。你当前的时间是 {Time}，你的位置是 {Location}。根据推荐的帖子列表（见下文），以 JSON 输出以下评估结果：1. 有多少帖子与您的生活圈相关（在 5 公里范围内或熟悉的社区）？2. 对每个帖子，给出一个匹配分数（0.0 到 1.0），表示内容对您的有用程度。3. 哪个位置（从 1 开始）是您觉得最有趣的帖子？输出格式（仅返回 JSON）：{ "relevant_post_count": <int>, "match_score": [<float>, <float>, ...], "top_interest_position": <int> }帖子信息: [{ "sname": "深圳大学城", "title": "跳蚤市场 | 二手自行车出售", "post_content": "在校园出售一辆几乎全新的自行车，八成新，诚心出价","}

"时间": "2024-03-01 14:52:00", "纬度": 22.590, "经度": 113.943, "标签": ["深圳", "南山", "校园", "二手"]